

Section 29 MAC(Media Access Controller)

29.1 Introduction

29.1.1 Overview

The MAC enables a host to transmit and receive data over Ethernet in compliance with the IEEE 802.3-2005 standard. You can use the MAC in number of applications such as switches and network interface cards.

29.1.2 MAC Features

The MAC has the following features:

- Supports 10/100 Mbps data transfer rates with the following PHY interfaces:
 - IEEE 802.3-compliant RMII interface to communicate with an external Fast Ethernet PHY.
- Provides the following features for full-duplex operation:
 - IEEE 802.3x flow control automatic transmission of zero-quanta pause frame on flow control input de-assertion.
 - Optional forwarding of received pause control frames to the user application.
- Provides the following features for half-duplex operation:
 - CSMA/CD Protocol support
 - Flow control using back-pressure support
 - Frame bursting and frame extension in 100 Mbps half-duplex operation
- Supports preamble and start-of-frame data (SFD) insertion in Transmit path. In the Receive path, the MAC core supports preamble and SFD deletion.
- Supports Automatic CRC and pad generation controllable on a per-frame basis.
- Provides options for Automatic Pad/CRC Stripping on receive frames.
- Supports a variety of flexible address filtering modes such as:
 - Up to 31 additional 48-bit perfect (DA) address filters with masks for each byte.
 - Up to 31 48-bit SA address comparison check with masks for each byte.

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- 64-bit Hash filter (optional) for multicast and unicast (DA) addresses.
- Option to pass all multicast addressed frames.
- Promiscuous mode support to pass all frames without any filtering for network monitoring.
- Passes all incoming packets (as per filter) with a status report.
- Supports programmable frame length to support Standard or Jumbo Ethernet frames with up to 16 KB of size.
- Supports programmable Interframe MACap (IFMAC) (40-96 bit times in steps of 8).
- Supports separate 32-bit status for transmit and receive packets.
- Supports IEEE 802.1Q VLAN tag detection for reception frames.
- Supports separate transmission, reception, and control interfaces to the Application.
- Supports configurable big-endian and little-endian for the Transmit and Receive paths.
- Provides the support for network statistics (optional) with RMON/MIB Counters (RFC2819/RFC2665).
- Provides a module that you can use for detection of LAN wake-up frames and AMD Magic Packet frames.
- Provides a Receive module for checksum off-load for received IPv4 and TCP packets encapsulated by the Ethernet frame (Type 1).
- Provides an optional Enhanced Receive module for checking IPv4 header checksum and TCP, UDP, or ICMP checksum encapsulated in IPv4 or IPv6 datagrams (Type 2).
- Supports MDIO Master interface (optional) for PHY device configuration and management.
- Supports the standard IEEE P802.3az, version D2.0 for Energy Efficient Ethernet.

29.1.3 System Overview

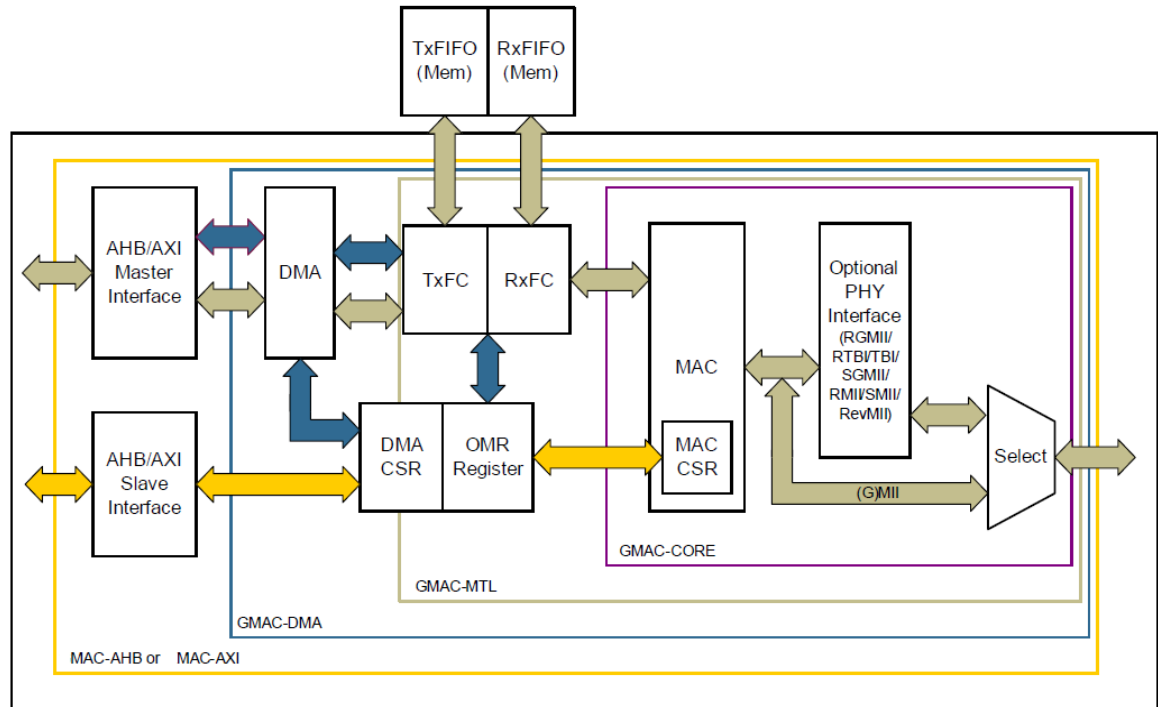


Figure 29-1 shows a system-level block diagram of MAC

The MAC transfers data to system memory through the AHB master interface. The host CPU uses the AHB Slave interface to access the MAC subsystem's Control and Status registers (CSRs).

29.1.3.1 Transmit and Receive FIFOs

The Transmit FIFO (Tx FIFO) buffers data read from system memory by the DMA before transmission by the MAC Core. Similarly, the Receive FIFO (Rx FIFO) stores the Ethernet frames received from the line until they are transferred to system memory by the DMA. These are asynchronous FIFOs, as they also transfer the data between the application clock and the MAC line clocks. The depth of this frame-length FIFO is 2K bytes for each Tx or Rx FIFO.

29.1.3.2 Transaction Layer(MTL)

The MTL block consists of two sets of FIFOs: a Transmit FIFO with programmable threshold capability, and a Receive FIFO with a programmable threshold (default of 64 bytes).

29.1.3.3 DMA Block Features

The DMA block exchanges data between the MTL block and host memory. The host can use a set of registers (DMA CSR) to control the DMA operations.

29.1.3.4 AMBA AHB Slave Interface Features

The AHB 32-bit Slave interface provides access to the DMA and MAC CSR space. The AHB Slave interface has the following characteristics:

- Fully AMBA 2.0 Compliant AHB Slave
- Supports single and all burst transfers.
- Supports busy and early terminations.
- MAC generates OKAY only response; does not generate SPLIT, RETRY, or ERROR responses.

29.1.3.5 AMBA AHB Master Interface Features

The DMA Controller interfaces with the Host through the AMBA 3 AHB Interface. The AMBA 3 AHB bus interface provides the characteristics to support highly effective data traffic throughput.

29.2 DMA controller

The DMA has independent Transmit and Receive engines, and a CSR space. The Transmit engine transfers data from system memory to the device port (MTL), while the Receive engine transfers data from the device port to the system memory. The controller uses descriptors to efficiently move data from source to destination with minimal Host CPU intervention. The DMA is designed for packet-oriented data transfers such as frames in Ethernet. The controller can be programmed to interrupt the Host CPU for situations such as Frame Transmit and Receive transfer completion, and other normal/error conditions.

The DMA and the Host driver communicate through two data structures:

- Control and Status registers (CSR)
- Descriptor lists and data buffers

Control and Status registers are described in detail in **29.5.2.1.6**. Descriptors are described in detail in **29.3**.

The DMA transfers data frames received by the core to the Receive Buffer in the Host memory, and Transmit data frames from the Transmit Buffer in the Host memory. Descriptors that reside in the Host memory act as pointers to these buffers.

There are two descriptor lists; one for reception, and one for transmission. The base address of each list is written into DMA Registers 3 and 4, respectively. A descriptor list is forward linked (either implicitly or explicitly). The last descriptor may point back to the first entry to create a ring structure. Explicit chaining of descriptors is accomplished by setting the second address chained in both Receive and Transmit descriptors (RDES1[24] and TDES1[24]). The descriptor lists reside in the Host physical memory address space. Each descriptor can point to a maximum of two buffers. This enables two buffers to be used, physically addressed, rather than contiguous buffers in memory.

A data buffer resides in the Host physical memory space, and consists of an entire frame or part of a frame, but cannot exceed a single frame. Buffers contain only data, buffer status is maintained in the descriptor. Data chaining refers to frames that span multiple data buffers. However, a single descriptor cannot span multiple frames. The DMA skips to the next frame buffer when end-of-frame is detected. Data chaining can be enabled or disabled.

The descriptor ring and chain structure is shown in

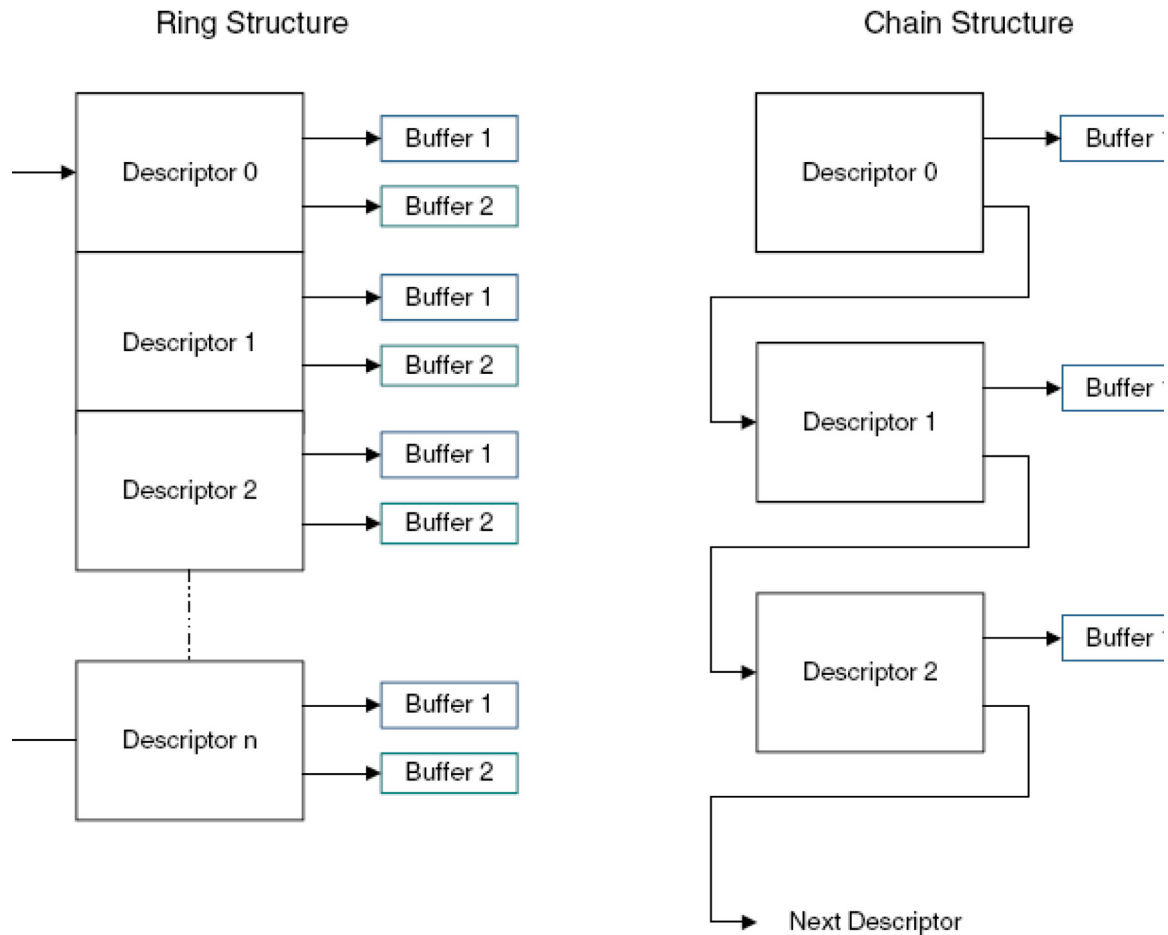


Figure 29-2 Descriptor Ring and Chain Structure

29.2.1 Initialization

Initialization for the MAC is as follows.

1. Write to DMA Register 0 to set Host bus access parameters.
2. Write to DMA Register 7 to mask unnecessary interrupt causes.
3. The software driver creates the Transmit and Receive descriptor lists. Then it writes to both DMA Register 3 and DMA Register 4, providing the DMA with the starting address of each list.
4. Write to MAC Registers 1, 2, and 3 for desired filtering options.
5. Write to MAC Register 0 to configure the operating mode and enable the

transmit operation (bit 3: Transmitter Enable). The PS and DM bits are set based on the auto-negotiation result (read from the PHY).

6. Write to DMA Register 6 to set bits 13 and 1 to start transmission and reception.
7. Write to MAC Register 0 to enable the Receive operation (bit 2: Receiver Enable).

The Transmit and Receive engines enter the Running state and attempt to acquire descriptors from the respective descriptor lists. The Receive and Transmit engines then begin processing Receive and Transmit operations. The Transmit and Receive processes are independent of each other and can be started or stopped separately.

29.2.1.1 Host Bus Burst Access

The DMA attempts to execute fixed-length Burst transfers on the AHB/AXI Master interface if configured to do so (FB bit of DMA Register 0). The maximum Burst length is indicated and limited by the PBL field (DMA Register 0[13:8]). The Receive and Transmit descriptors are always accessed in the maximum possible (limited by PBL or $16 * 8/\text{bus width}$) burst-size for the 16-bytes to be read.

The Transmit DMA initiates a data transfer only when sufficient space to accommodate the configured burst is available in MTL Transmit FIFO or the number of bytes till the end of frame (when it is less than the configured burst-length). The DMA indicates the start address and the number of transfers required to the AHB/AXI Master Interface. When the AHB/AXI Interface is configured for fixed-length burst, then it transfers data using the best combination of INCR4/8/16 and SINMACLE transactions. Otherwise (no fixedlength burst), it transfers data using INCR (undefined length) and SINMACLE transactions.

The Receive DMA initiates a data transfer only when sufficient data to accommodate the configured burst is available in MTL Receive FIFO or when the end of frame (when it is less than the configured burst-length) is detected in the Receive FIFO. The DMA indicates the start address and the number of transfers required to the AHB/AXI Master Interface. When the AHB/AXI Interface is configured for fixed-length burst, then it transfers data using the best combination of INCR4/8/16 and SINMACLE transactions. If the end-of frame is reached before the fixed-burst ends on the AHB/AXI interface, then dummy transfers are performed inorder to complete the fixed-burst. Otherwise (FB bit of DMA Register 0 is reset), it transfers data using INCR (undefined length) and SINMACLE transactions.

When the AHB/AXI interface is configured for address-aligned beats, both DMA engines ensure that the first burst transfer the AHB/AXI initiates is less than or equal to the size of the configured PBL. Thus, all subsequent beats start at an address that is aligned to the configured PBL. The DMA can only align the address for beats up to size 16 (for PBL > 16), because the AHB/AXI interface does not support more than INCR16.

29.2.1.2 Host Data Buffer Alignment

The Transmit and Receive data buffers do not have any restrictions on start address alignment. For example, in systems with 32-bit memory, the start address for the buffers can be aligned to any of the four bytes. However, the DMA always initiates transfers with address aligned to the bus width with dummy data for the byte lanes not required. This typically happens during the transfer of the beginning or end of an Ethernet frame.

Example Buffer Read

If the Transmit buffer address is 32'h00000FF2 (for 32-bit data bus), and 15 bytes need to be transferred, then the DMA reads five full words from address 32'h00000FF0, but when transferring data to the MTL Transmit FIFO, the extra bytes (the first two bytes) are dropped or ignored. Similarly, the last 3 bytes of the last transfer are also ignored. The DMA always ensures it transfers a full 32-bit data to the MTL Transmit FIFO, unless it is the end-of-frame.

Example Buffer Write

If the Receive buffer address is 32'h00000FF2 (for 64-bit data bus) and 16 bytes of a received frame need to be transferred, then the DMA writes 3 full words from address 32'h00000FF0. But the first 2 bytes of first transfer and the last 6 bytes of the third transfer have dummy data.

29.2.1.3 Buffer Size Calculations

The DMA does not update the size fields in the Transmit and Receive descriptors. The DMA updates only the status fields (RDES and TDES) of the descriptors. The driver has to perform the size calculations.

The transmit DMA transfers the exact number of bytes (indicated by buffer size field of TDES1) towards the MAC core. If a descriptor is marked as first (FS bit of TDES1 is set), then the DMA marks the first transfer from the buffer as the start of frame. If a descriptor is marked as last (LS bit of TDES1), then the DMA marks the last transfer from that data buffer as the end-of frame to the MTL.

The Receive DMA transfers data to a buffer until the buffer is full or the end-of frame is received from the MTL. If a descriptor is not marked as last (LS bit of RDES0), then the descriptor's corresponding buffer(s) are full and the amount of valid data in a buffer is accurately indicated by its buffer size field minus the data buffer pointer offset when the FS bit of that descriptor is set. The offset is zero when the data buffer pointer is aligned to the data bus width. If a descriptor is marked as last, then the buffer may not be full (as indicated by the buffer size in RDES1). To compute the amount of valid data in this final buffer, the driver must read the frame length (FL bits of RDES0[29:16]) and subtract the sum of the buffer sizes of the preceding buffers in this frame. The Receive DMA always transfers the start of next frame with a new descriptor.

Note : Even when the start address of a receive buffer is not aligned to the system bus's data width, the system should allocate a receive buffer of a size aligned to the system bus width. For example, if the system allocates a 1,024-byte (1 KB) receive buffer starting from address 0x1000, the software can program the buffer start address in the Receive descriptor to have a 0x1002 offset. The Receive DMA writes the frame to this buffer with dummy data in the first two locations (0x1000 and 0x1001). The actual frame is written from location 0x1002. Thus, the actual useful space in this buffer is 1,022 bytes, even though the buffer size is programmed as 1,024 bytes, because of the start address offset.

29.2.1.4 DMA Arbiter

The arbiter inside the DMA module performs the arbitration between the Transmit and Receive channel accesses to the AHB Master interface. Two types of arbitrations are possible: round-robin, and fixedpriority.

When round-robin arbitration is selected (DA bit of Register 0 (Bus Mode Register) is reset), the arbiter allocates the data bus in the ratio set by the PR bits of DMA Register 0, when both Transmit and Receive DMAs are requesting for access simultaneously. When the DA bit is set, the Receive DMA always gets priority over the Transmit DMA for data access by default. When the TXPR bit (bit 27 of DMA register 0) is also set, then the Transmit DMA gets priority over the Receive DMA as explained in **Table 29-1**

Table 29-1 Priority Scheme for Tx DMA and Rx DMA

Bit 27	Bit 15	Bit 14	Bit 1	Priority Scheme
0	x	x	1	Rx always has priority over Tx
0	0	0	0	Tx and Rx have equal priority. Rx gets the access first on simultaneous requests.
0	0	1	0	Rx has priority over Tx in the ratio 2:1.
0	1	0	0	Rx has priority over Tx in the ratio 3:1.
0	1	1	0	Rx has priority over Tx in the ratio 4:1.
1	x	x	1	Tx always has priority over Rx.
1	0	0	0	Tx and Rx have equal priority. Tx gets the access first on simultaneous requests.
1	0	1	0	Tx has priority over Rx in the ratio 2:1.
1	1	0	0	Tx has priority over Rx in the ratio 3:1.
1	1	1	0	Tx has priority over Rx in the ratio 4:1.

29.2.2 Transmission

29.2.2.1 TxDMA Operation :Default(Non-OSF)Mode

The Transmit DMA engine in default mode proceeds as follows:

1. The Host sets up the transmit descriptor (TDES0-TDES3) and sets the Own bit (TDES0[31]) after setting up the corresponding data buffer(s) with Ethernet Frame data.
2. Once the ST bit (DMA Register 6[13]) is set, the DMA enters the Run state.
3. While in the Run state, the DMA polls the Transmit Descriptor list for frames requiring transmission. After polling starts, it continues in either sequential descriptor ring order or chained order. If the DMA detects a descriptor flagged as owned by the Host, or if an error condition occurs, transmission is suspended and both the Transmit Buffer Unavailable (DMA Register 5[2]) and Normal Interrupt Summary (DMA Register 5[16]) bits are set. The Transmit Engine proceeds to Step 9.

4. If the acquired descriptor is flagged as owned by DMA (TDES0[31] = 1'b1), the DMA decodes the Transmit Data Buffer address from the acquired descriptor.
5. The DMA fetches the Transmit data from the Host memory and transfers the data to the MTL for transmission.
6. If an Ethernet frame is stored over data buffers in multiple descriptors, the DMA closes the intermediate descriptor and fetches the next descriptor. Steps 3, 4, and 5 are repeated until the end-of-Ethernet-frame data is transferred to the MTL.
7. When frame transmission is complete, if IEEE 1588 time stamping was enabled for the frame (as indicated in the transmit status) the timestamp value obtained from MTL is written to the transmit descriptor (TDES2 and TDES3) that contains the end-of-frame buffer. The status information is then written to this transmit descriptor (TDES0). Because the Own bit is cleared during this step, the Host now owns this descriptor. If time stamping was not enabled for this frame, the DMA does not alter the contents of TDES2 and TDES3.
8. Transmit Interrupt (DMA Register 5[0]) is set after completing transmission of a frame that has Interrupt on Completion (TDES1[31]) set in its Last Descriptor. The DMA engine then returns to Step 3.
9. In the Suspend state, the DMA tries to re-acquire the descriptor (and thereby return to Step 3) when it receives a Transmit Poll demand and the Underflow Interrupt Status bit is cleared.

The TxDMA transmission flow in default mode is shown in **Figure 29-3**

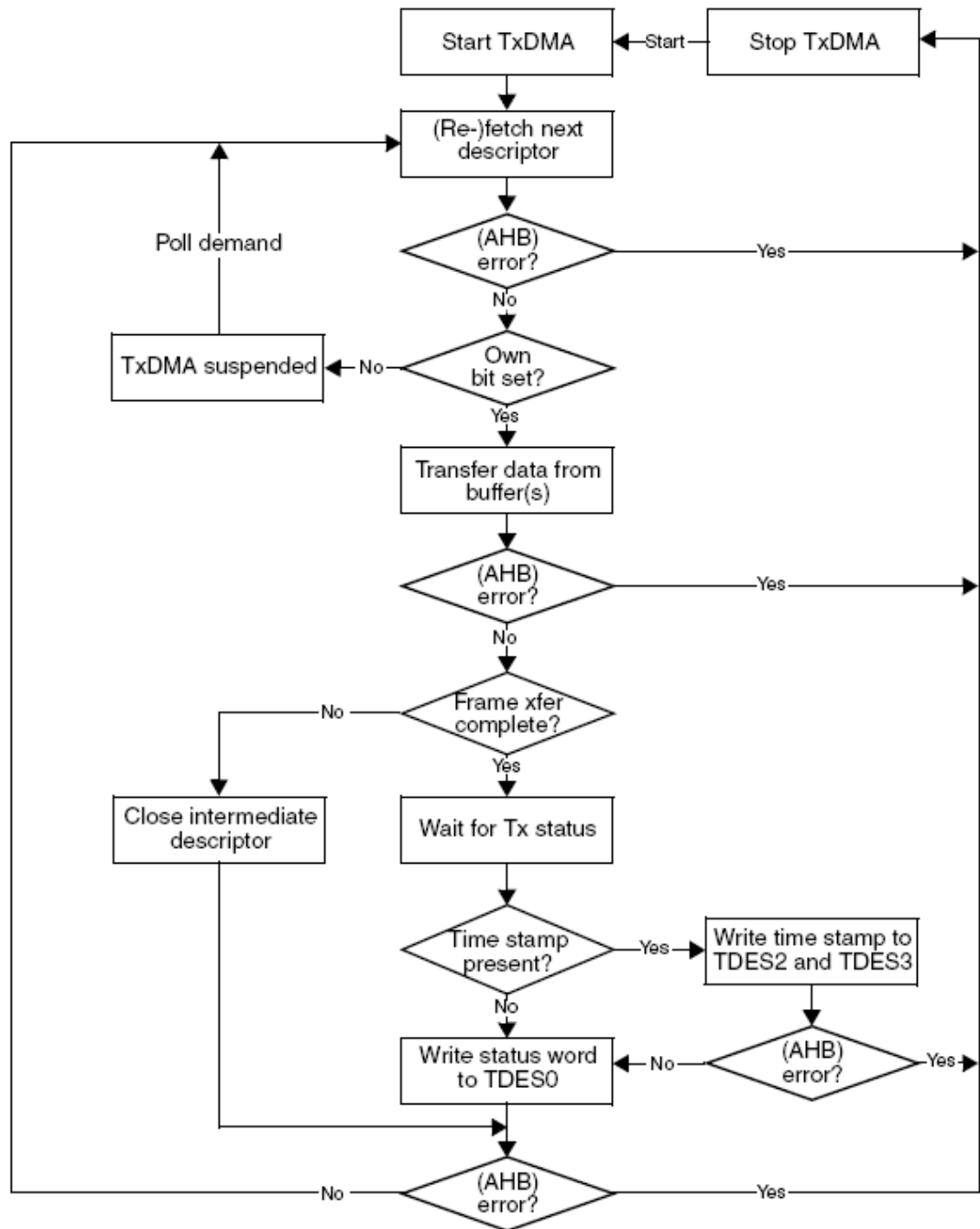


Figure 29-3 TxDMA Operation in Default Mode

29.2.2.2 TxDMA Operation :OSF Mode

While in the Run state, the transmit process can simultaneously acquire two frames without closing the Status descriptor of the first (if the OSF bit is set in DMA Register 6[2]). As the transmit process finishes transferring the first frame, it immediately polls the Transmit Descriptor list for the second frame. If the second frame is valid, the transmit process transfers this frame before writing the first frame's status information.

In OSF mode, the Run state Transmit DMA operates in the following sequence:

1. The DMA operates as described in steps 1~6 of the TxDMA (default mode).
2. Without closing the previous frame's last descriptor, the DMA fetches the next descriptor.
3. If the DMA owns the acquired descriptor, the DMA decodes the transmit buffer address in this descriptor. If the DMA does not own the descriptor, the DMA goes into Suspend mode and skips to Step 7.
4. The DMA fetches the Transmit frame from the Host memory and transfers the frame to the MTL until the End-of-Frame data is transferred, closing the intermediate descriptors if this frame is split across multiple descriptors.
5. The DMA waits for the previous frame's frame transmission status and time stamp. Once the status is available, the DMA writes the time stamp to TDES2 and TDES3, if such time stamp was captured (as indicated by a status bit). The DMA then writes the status, with a cleared Own bit, to the corresponding TDES0, thus closing the descriptor. If time stamping was not enabled for the previous frame, the DMA does not alter the contents of TDES2 and TDES3.
6. If enabled, the Transmit interrupt is set, the DMA fetches the next descriptor, then proceeds to Step 3 (when Status is normal). If the previous transmission status shows an underflow error, the DMA goes into Suspend mode (Step 7).
7. In Suspend mode, if a pending status and time stamp are received from the MTL, the DMA writes the time stamp (if enabled for the current frame) to TDES2 and TDES3, then writes the status to the corresponding TDES0. It then sets relevant interrupts and returns to Suspend mode.
8. The DMA can exit Suspend mode and enter the Run state (go to Step 1 or Step 2 depending on pending status) only after receiving a Transmit Poll demand (DMA Register 1).

Note:As the DMA fetches the next descriptor in advance before closing the current descriptor, the descriptor chain should have more than 2 different descriptors for correct and proper operation.

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The basic flow is described in **Figure 29-4**

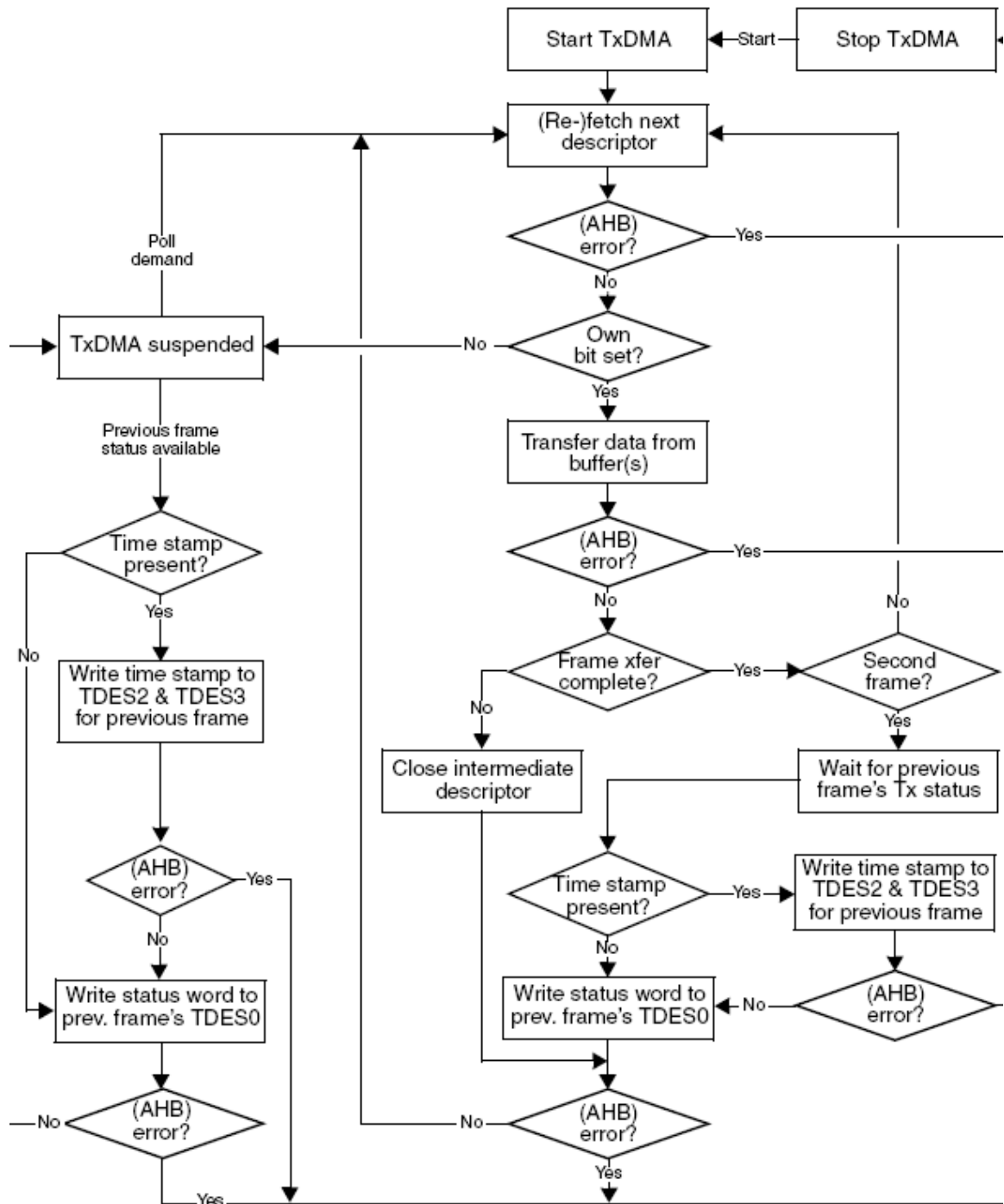


Figure 29-4 TxDMA Operation in OSF Mode

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29.2.2.3 Transmit Frame Processing

The Transmit DMA expects that the data buffers contain complete Ethernet frames, excluding preamble, pad bytes, and FCS fields. The DA, SA, and Type/Len fields contain valid data. If the Transmit Descriptor indicates that the MAC core must disable CRC or PAD insertion, the buffer must have complete Ethernet frames (excluding preamble), including the CRC bytes.

Frames can be data-chained and can span several buffers. Frames must be delimited by the First Descriptor (TDES1[29]) and the Last Descriptor (TDES1[30]), respectively.

As transmission starts, the First Descriptor must have (TDES1[29]) set. When this occurs, frame data transfers from the Host buffer to the MTL Transmit FIFO. Concurrently, if the current frame has the Last Descriptor (TDES1[30]) clear, the Transmit Process attempts to acquire the Next Descriptor. The Transmit Process expects this descriptor to have TDES1[29] clear. If TDES1[30] is clear, it indicates an intermediary buffer. If TDES1[30] is set, it indicates the last buffer of the frame.

After the last buffer of the frame has been transmitted, the DMA writes back the final status information to the Transmit Descriptor 0 (TDES0) word of the descriptor that has the last segment set in Transmit Descriptor 1 (TDES1[30]). At this time, if Interrupt on Completion (TDES1[31]) was set, Transmit Interrupt (DMA Register 5[0]) is set, the Next Descriptor is fetched, and the process repeats.

The actual frame transmission begins after the MTL Transmit FIFO has reached either a programmable transmit threshold (DMA Register 6[16:14]), or a full frame is contained in the FIFO. There is also an option for Store and Forward Mode (DMA Register 6[21]). Descriptors are released (Own bit TDES0[31] clears) when the DMA finishes transferring the frame.

Note: To ensure proper transmission of a frame and the next frame, you must specify a non-zero buffer size for the transmit descriptor that has the Last Descriptor (TDES1[30]) set.

29.2.2.4 Transmit Polling Suspended

Transmit polling can be suspended by either of the following conditions:

- The DMA detects a descriptor owned by the Host (TDES0[31]=0). To resume, the driver must give descriptor ownership to the DMA and then issue a Poll Demand command.
- A frame transmission is aborted when a transmit error because of underflow is detected. The appropriate Transmit Descriptor 0 (TDES0) bit is set.

If the second condition occur, both Abnormal Interrupt Summary (DMA Register 5[15]) and Transmit Underflow bits (DMA Register 5 [5]) are set, and the information is written to Transmit Descriptor 0, causing the suspension. If the DMA goes into SUSPEND state because of the first condition, then both Normal Interrupt Summary (DMA Register 5 [16]) and Transmit Buffer Unavailable (DMA Register 5 [2]) are set.

In both cases, the position in the Transmit List is retained. The retained position is that of the descriptor following the Last Descriptor closed by the DMA.

The driver must explicitly issue a Transmit Poll Demand command after rectifying the suspension cause.

29.2.3 Reception

The Receive DMA engine's reception sequence is depicted in **Figure 29-5** and proceeds as follows:

1. The host sets up Receive descriptors (RDES0-RDES3) and sets the Own bit (RDES0[31]).
2. Once the SR (DMA Register 6[1]) bit is set, the DMA enters the Run state. While in the Run state, the DMA polls the Receive Descriptor list, attempting to acquire free descriptors. If the fetched descriptor is not free (is owned by the host), the DMA enters the Suspend state and jumps to Step 9.
3. The DMA decodes the receive data buffer address from the acquired descriptors.
4. Incoming frames are processed and placed in the acquired descriptor's data buffers.
5. When the buffer is full or the frame transfer is complete, the Receive engine fetches the next descriptor.
6. If the current frame transfer is complete, the DMA proceeds to Step 7. If the DMA does not own the next fetched descriptor and the frame transfer is not complete (EOF is not yet transferred), the DMA sets the Descriptor Error bit in the RDES0 (unless flushing is disabled). The DMA closes the current descriptor (clears the Own bit) and marks it as intermediate by clearing the Last Segment (LS) bit in the RDES0 value (marks it as Last Descriptor if flushing is not disabled), then proceeds to Step 8. If the DMA does own the next descriptor but the current frame transfer is not complete, the DMA closes the current descriptor as intermediate and reverts to Step 4.
7. If IEEE 1588 time stamping is enabled, the DMA writes the timestamp (if available) to the current descriptor's RDES2 and RDES3. It then takes the

receive frame's status from the MTL and writes the status word to the current descriptor's RDES0, with the Own bit cleared and the Last Segment bit set.

8. The Receive engine checks the latest descriptor's Own bit. If the host owns the descriptor (Own bit is 1'b0) the Receive Buffer Unavailable bit (Register 5[7]) is set and the DMA Receive engine enters the Suspended state (Step 9). If the DMA owns the descriptor, the engine returns to Step 4 and awaits the next frame.
9. Before the Receive engine enters the Suspend state, partial frames are flushed from the Receive FIFO (You can control flushing using Bit 24 of DMA Register 6).
10. The Receive DMA exits the Suspend state when a Receive Poll demand is given or the start of next frame is available from the MTL's Receive FIFO. The engine proceeds to Step 2 and refetches the next descriptor.

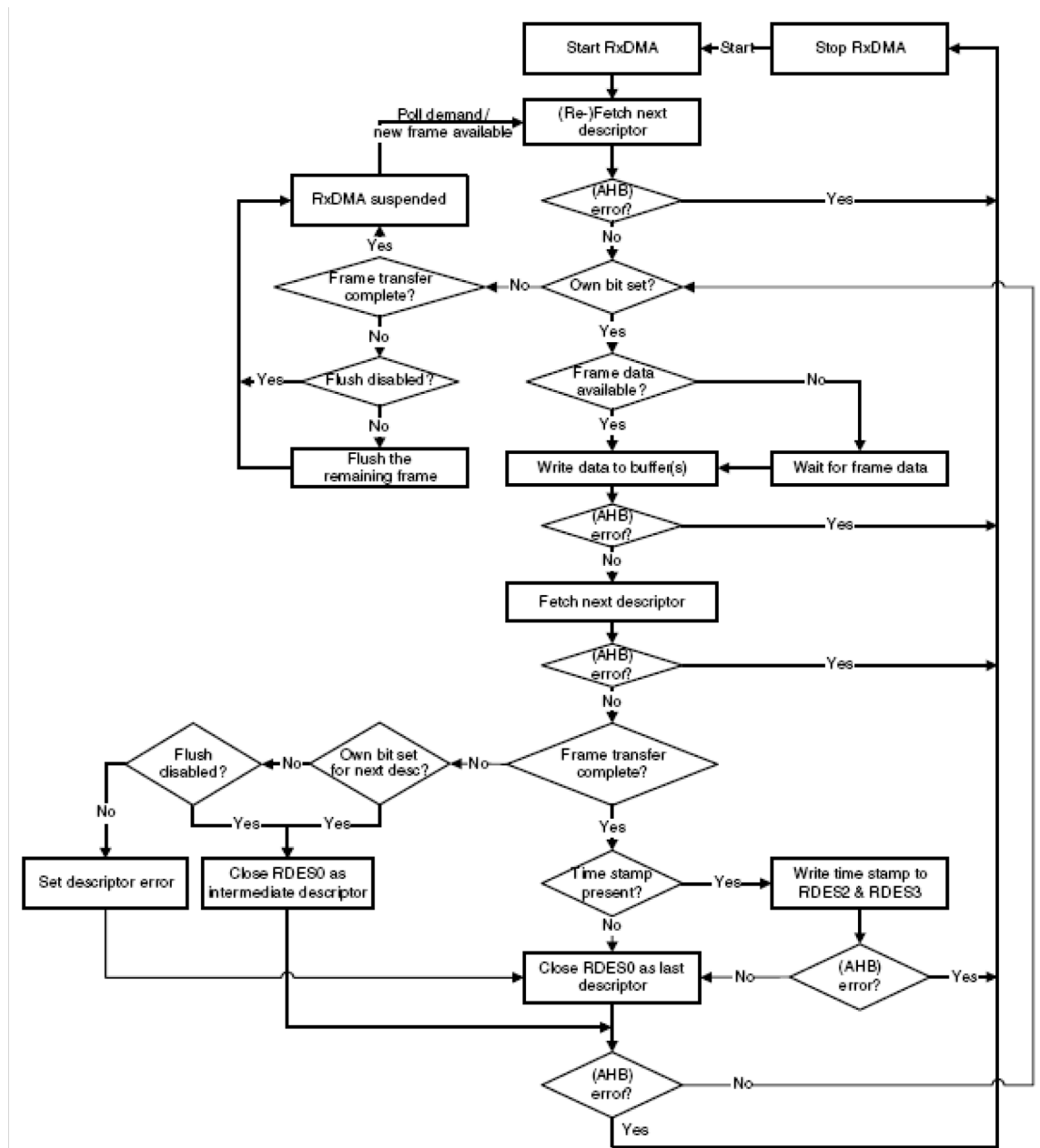


Figure 29-5 Receive DMA Operation

The DMA does not acknowledge accepting the status from the MTL until it has completed the time stamp write-back and is ready to perform status write-back to the descriptor.

If software has enabled time stamping through CSR, when a valid time stamp value is not available for the frame (for example, because the receive FIFO was full before the time stamp could be written to it), the DMA writes all-ones to RDES2 and RDES3. Otherwise (that is, if time stamping is not enabled), the RDES2 and RDES3 remain unchanged.

29.2.3.1 Receive Descriptor Acquisition

The Receive Engine always attempts to acquire an extra descriptor in anticipation of an incoming frame. Descriptor acquisition is attempted if any of the following conditions is satisfied:

- The receive Start/Stop bit (Register 6[1]) has been set immediately after being placed in the Run state.
- The data buffer of current descriptor is full before the frame ends for the current transfer.
- The controller has completed frame reception, but the current Receive Descriptor is not yet closed.
- The receive process has been suspended because of a host-owned buffer (RDES0[31] = 0) and a new frame is received.
- A Receive poll demand has been issued.

29.2.3.2 Receive Frame Processing

The MAC transfers the received frames to the Host memory only when the frame passes the address filter and frame size is greater than or equal to configurable threshold bytes set for the Receive FIFO of MTL, or when the complete frame is written to the FIFO in Store-and-Forward mode.

If the frame fails the address filtering, it is dropped in the MAC block itself (unless Receive All MAC Register 1[31] bit is set). Frames that are shorter than 64 bytes, because of collision or premature termination, can be purged from the MTL Receive FIFO.

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The DMA fetches the next descriptor, sets the Last Descriptor (RDES[8]) bit, and releases the RDES0 status bits in the previous frame descriptor. Then the DMA sets Receive Interrupt (Register 5[6]). The same process repeats unless the DMA

encounters a descriptor flagged as being owned by the host. If this occurs, the Receive Process sets Receive Buffer Unavailable (Register 5[7]) and then enters the Suspend state. The position in the receive list is retained.

29.2.3.3 Receive Process Suspended

If a new Receive frame arrives while the Receive Process is in Suspend state, the DMA refetches the current descriptor in the Host memory. If the descriptor is now owned by the DMA, the Receive Process re-enters the Run state and starts frame reception. If the descriptor is still owned by the host, by default, the DMA discards the current frame at the top of the MTL Rx FIFO and increments the missed frame counter. If more than one frame is stored in the MTL Rx FIFO, the process repeats.

The discarding or flushing of the frame at the top of the MTL Rx FIFO can be avoided by setting Operation Mode register bit 24 (DFF). In such conditions, the receive process sets the Receive Buffer Unavailable status and returns to the Suspend state.

29.2.4 Interrupt

Interrupts can be generated as a result of various events. The DMA Register 5 **29.5.2.1.6**(Status Register) contains all the bits that might cause an interrupt. Register 7 **29.5.2.1.8**(Interrupt Enable Register) contains an enable bit for each of the events that can cause an interrupt.

There are two groups of interrupts, Normal and Abnormal, as described in Register 5 (Status Register). Interrupts are cleared by writing a 1'b1 to the corresponding bit position. When all the enabled interrupts within a group are cleared, the corresponding summary bit is cleared. When both the summary bits are cleared, the interrupt signal `sbd_intr_o` is de-asserted. If the MAC core is the cause for assertion of the interrupt, then any of the MACLI, MACMI, or MACPI bits of DMA Register 5 (Status Register) are set high.

Note:The Register 5 (Status Register) is the (interrupt) status register. The interrupt pin (`sbd_intr_o`) is asserted because of any event in this status register only if the corresponding interrupt enable bit is set in Register 7 (Interrupt Enable Register).

Interrupts are not queued and if the interrupt event occurs before the driver has responded to it, no additional interrupts are generated. For example, Receive Interrupt (bit 6 of Register 5 (Status Register)) indicates that one or more frames were transferred to the Host buffer. The driver must scan all descriptors, from the last recorded position to the first one owned by the DMA.

An interrupt is generated only once for simultaneous, multiple events. The driver must scan the Register 5 (Status Register) for the cause of the interrupt. The interrupt is not generated again unless a new interrupting event occurs, after the driver has cleared the appropriate bit in Register 5 (Status Register). For example, the controller generates a Receive interrupt (bit 6 of Register 5 (Status Register)) and the driver begins reading Register 5 (Status Register). Next, Receive Buffer Unavailable (bit 7 of Register 5 (Status Register)) occurs. The driver clears the Receive interrupt. Even then, the `sbd_intr_o` signal is not de-asserted, because of the active or pending Receive Buffer Unavailable interrupt.

An interrupt timer RIWT (bits 7:0 in Register 9 **29.5.2.1.10**(Receive Interrupt Watchdog Timer Register)) is given for flexible control of Receive Interrupt. When this Interrupt timer is programmed with a non-zero value, it gets activated as soon as the RxDMA completes a transfer of a received frame to system memory without asserting the Receive Interrupt because it is not enabled in the corresponding Receive Descriptor (RDES1[31] in **29.3.1.2**). When this timer runs out as per the programmed value, RI bit is set and the interrupt is asserted if the corresponding RI is enabled in Register 7 (Interrupt Enable Register). This timer gets disabled before it runs out, when a frame is transferred to memory and the RI is set because it is enabled for that descriptor.

29.2.5 Error Response DMA

For any data transfer initiated by a DMA channel, if the slave replies with an error response, that DMA stops all operations and updates the error bits and the Fatal Bus Error bit in the Status register (DMA Register 5). That DMA controller can resume operation only after soft resetting or hard resetting the core and re-initializing the DMA.

29.3 Descriptor

29.3.1 Receive Descriptor

The MAC Subsystem requires at least two descriptors when receiving a frame. The Receive state machine of the DMA (in the MAC Subsystem) always attempts to acquire an extra descriptor in anticipation of an incoming frame. (The size of the incoming frame is unknown). Before the RxDMA closes a descriptor, it attempts to acquire the next descriptor even if no frames are received.

In a single descriptor (receive) system, the subsystem generates a descriptor error if the receive buffer is unable to accommodate the incoming frame and the next descriptor is not owned by the DMA. Thus, the Host is forced to increase either its

descriptor pool or the buffer size. Otherwise, the subsystem starts dropping all incoming frames.

For continuous rx data mode, rx descriptors must work in "second address chained" mode, and the addresses of RDES1 and RDES2 exchange. Buffer1 base address in RDES2 is useless. Rx DMA writes frame_length and frame base address back to both RDES0 and RDES2. The buffer1 size (byte count buffer 1) should be configured large enough to avoid big frame crossing several descriptors.

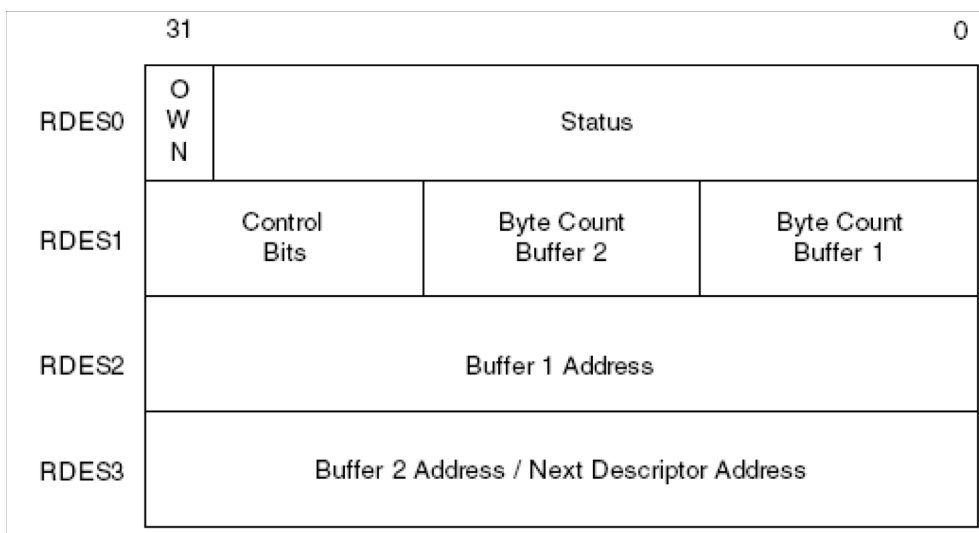


Figure 29-6 Receive Descriptor Format

29.3.1.1 Receive Descriptor 0(RDES0)

RDES0 contains the received frame status, the frame length, and the descriptor ownership information. The descriptions in the following tables are for the default mode of a little-endian 32-bit data bus with Same Endian descriptors, or big-endian data bus with Reverse-Endian descriptors. If the DUT is configured for big-endian mode, 32-bit data bus with Same Endian descriptors or little-endian data bus with Reverse Endian descriptors, then byte lanes 3 and 0 are swapped while byte lanes 1 and 2 are swapped, that is, bits [31:24] are available on data bus [7:0], and vice-versa.

Table 29-2 Receive Descriptor 0

Bit	Description
31	OWN: Own Bit When set, this bit indicates that the descriptor is owned by the DMA of the MAC Subsystem. When this bit is reset, this bit indicates that the descriptor is owned by the Host. The DMA clears this bit either when it completes the frame reception or when the buffers that are associated with this descriptor are full.
30	AFM: Destination Address Filter Fail When set, this bit indicates a frame that failed in the DA Filter in the MAC Core.
29:16	FL: Frame Length These bits indicate the byte length of the received frame that was transferred to host memory (including CRC). This field is valid when Last Descriptor (RDES0[8]) is set and either the Descriptor Error (RDES0[14]) or Overflow Error bits are reset. The frame length also includes the two bytes appended to the Ethernet frame when IP checksum calculation (Type 1) is enabled and the received frame is not a MAC control frame. This field is valid when Last Descriptor (RDES0[8]) is set. When the Last Descriptor and Error Summary bits are not set, this field indicates the accumulated number of bytes that have been transferred for the current frame.
15	ES: Error Summary Indicates the logical OR of the following bits: RDES0[0]: Payload Checksum Error RDES0[1]: CRC Error RDES0[3]: Receive Error RDES0[4]: Watchdog Timeout RDES0[6]: Late Collision RDES0[7]: IPC Checksum (Type 2) / MACiant Frame RDES0[11]: Overflow Error RDES0[14]: Descriptor Error This field is valid only when the Last Descriptor (RDES0[8]) is set.
14	DE: Descriptor Error When set, this bit indicates a frame truncation caused by a frame that does not fit within the current descriptor buffers, and that the DMA does not own the Next Descriptor. The frame is truncated. This field is valid only when the Last Descriptor (RDES0[8]) is set.
13	SAF: Source Address Filter Fail When set, this bit indicates that the SA field of frame failed the SA Filter in the MAC Core.
12	LE: Length Error When set, this bit indicates that the actual length of the frame received and that the Length/ Type field does not match. This bit is valid only when the Frame Type (RDES0[5]) bit is reset. Length error status is not valid when CRC error is present.
11	OE: Overflow Error When set, this bit indicates that the received frame was damaged due to buffer overflow in MTL.
10	VLAN: VLAN Tag When set, this bit indicates that the frame pointed to by this descriptor is a VLAN frame tagged by the MAC Core.
9	FS: First Descriptor When set, this bit indicates that this descriptor contains the first buffer of the frame. If the size of the first buffer is 0, the second buffer contains the beginning of the frame. If the size of the second buffer is also 0, the next Descriptor contains the beginning of the frame.
8	LS: Last Descriptor When set, this bit indicates that the buffers pointed to by this descriptor are the last buffers of the frame

Table 29-2 Receive Descriptor 0

Bit	Description
7	IPC Checksum Error/MACiant Frame When IP Checksum Engine (Type 1) is enabled, this bit, when set, indicates that the 16-bit IPv4 Header checksum calculated by the core did not match the received checksum bytes. The Error Summary bit[15] is NOT set when this bit is set in this mode. If this bit is set when Full Checksum Offload Engine (Type 2) is enabled, it indicates an error in the IPv4 or IPv6 header. This error can be due to inconsistent Ethernet Type field and IP header Version field values, a header checksum mismatch in IPv4, or an Ethernet frame lacking the expected number of IP header bytes. Refer to Table 29-3 for more details. If you do not select IP Checksum Module during core configuration, this bit, when set, indicates that the received frame was a MACiant Frame. MACiant frames are larger-than-1,518-byte (or 1,522-byte for VLAN) normal frames and larger-than-9,018-byte (9,022-byte for VLAN) frame when Jumbo Frame processing is enabled.
6	LC: Late Collision When set, this bit indicates that a late collision has occurred while receiving the frame in Half-Duplex mode.
5	FT: Frame Type When set, this bit indicates that the Receive Frame is an Ethernet-type frame (the LT field is greater than or equal to 16'h0600). When this bit is reset, it indicates that the received frame is an IEEE802.3 frame. This bit is not valid for Runt frames less than 14 bytes. Refer to Table 29-3 for more details.
4	RWT: Receive Watchdog Timeout When set, this bit indicates that the Receive Watchdog Timer has expired while receiving the current frame and the current frame is truncated after the Watchdog Timeout.
3	RE: Receive Error When set, this bit indicates that the gmii_rxr_i signal is asserted while gmii_rxdv_i is asserted during frame reception. This error also includes carrier extension error in MACMII and Half-duplex mode. Error can be of less/no extension, or error (rxd ≠ 0f) during extension.
2	DE: Dribble Bit Error When set, this bit indicates that the received frame has a non-integer multiple of bytes (odd nibbles). This bit is valid only in MII Mode.
1	CE: CRC Error When set, this bit indicates that a Cyclic Redundancy Check (CRC) Error occurred on the received frame. This field is valid only when the Last Descriptor (RDES0[8]) is set.
0	Rx MAC Address/Payload Checksum Error When set, this bit indicates that the Rx MAC Address registers value (1 to 15) matched the frame's DA field. When reset, this bit indicates that the Rx MAC Address Register 0 value matched the DA field. If Full Checksum Offload Engine is enabled, this bit, when set, indicates the TCP, UDP, or ICMP checksum the core calculated does not match the received encapsulated TCP, UDP, or ICMP segment's Checksum field. This bit is also set when the received number of payload bytes does not match the value indicated in the Length field of the encapsulated IPv4 or IPv6 datagram in the received Ethernet frame. Refer to Table 29-3 for more details.

When the Full Checksum Offload Engine (Type 2) is enabled, the permutations of bits 5, 7, and 0 reflect the conditions discussed in **Table 29-3**.

Table 29-3 Receive Descriptor 0 when COE(Type 2) Is Enable

Bit 5: Frame Type	Bit 7: IPC Checksum Error	Bit 0: Payload Checksum Error	Frame Status
0	0	0	IEEE 802.3 Type frame (Length field value is less than 0x0600.)

Table 29-3 Receive Descriptor 0 when COE(Type 2) Is Enable

Bit 5: Frame Type	Bit 7: IPC Checksum Error	Bit 0: Payload Checksum Error	Frame Status
1	0	0	IPv4/IPv6 Type frame, no checksum error detected
1	0	1	IPv4/IPv6 Type frame with a payload checksum error (as described for PCE) detected
1	1	0	IPv4/IPv6 Type frame with an IP header checksum error (as described for IPC CE) detected
1	1	1	IPv4/IPv6 Type frame with both IP header and payload checksum errors detected
0	0	1	IPv4/IPv6 Type frame with no IP header checksum error and the payload check bypassed, due to an unsupported payload
0	1	1	A Type frame that is neither IPv4 or IPv6 (the Checksum Offload engine bypasses checksum completely.)
0	1	0	Reserved

Note: The first five conditions are backward-compatible to versions 3.30a and previous. The last two conditions (001, 011), which had been reserved, are not backward-compatible, because the Frame Type (FT) bit is reset during bypasses, even for valid Type frames.

29.3.1.2 Receive Descriptor 1(RDES1)

RDES1 contains the buffer sizes and other bits that control the descriptor chain/ring.

Table 29-4 Receive Descriptor

Bit	Description
31	Disable Interrupt on Completion When set, this bit prevents the setting of the RI (CSR5[6]) bit of the Status Register for the received frame that ends in the buffer pointed to by this descriptor. This, in turn, disables the assertion of the interrupt to Host due to RI for that frame.
30:26	Reserved
25	RER: Receive End of Ring When set, this bit indicates that the descriptor list reached its final descriptor. The DMA returns to the base address of the list, creating a Descriptor Ring.
24	RCH: Second Address Chained When set, this bit indicates that the second address in the descriptor is the Next Descriptor address rather than the second buffer address. When RDES1[24] is set, RBS2 (RDES1[21-11]) is a “don’t care” value. RDES1[25] takes precedence over RDES1[24].
23:22	Reserved

Table 29-4 Receive Descriptor

Bit	Description
21:11	RBS2: Receive Buffer 2 Size These bits indicate the second data buffer size in bytes. The buffer size must be a multiple of 4/8/16 depending upon the bus widths (32/64/128), even if the value of RDES3 (buffer2 address pointer) is not aligned to bus width. In the case where the buffer size is not a multiple of 4/8/16, the resulting behavior is undefined. This field is not valid if RDES1[24] is set.
10:0	RBS1: Receive Buffer 1 Size Indicates the first data buffer size in bytes. The buffer size must be a multiple of 4/8/16 depending upon the bus widths (32/64/128), even if the value of RDES2 (buffer1 address pointer) is not aligned. In the case where the buffer size is not a multiple of 4/8/16, the resulting behavior is undefined. If this field is 0, the DMA ignores this buffer and uses Buffer 2 or next descriptor depending on the value of RCH (Bit 24).

29.3.1.3 Receive Descriptor 2(RDES2)

RDES2 contains the address pointer to the first data buffer in the descriptor.

Table 29-5 Receive Descriptor 2(Default Operation)

Bit	Description
31:0	Buffer 1 Address Pointer These bits indicate the physical address of Buffer 1. There are no limitations on the buffer address alignment except for the following condition: The DMA uses the configured value for its address generation when the RDES2 value is used to store the start of frame. Note that the DMA performs a write operation with the RDES2[3/2/1:0] bits as 0 during the transfer of the start of frame but the frame data is shifted as per the actual Buffer address pointer. The DMA ignores RDES2[3/2/1:0] (corresponding to bus width of 128/64/32) if the address pointer is to a buffer where the middle or last part of the frame is stored.

29.3.1.4 Receive Descriptor 3(RDES3)

RDES3 contains the address pointer either to the second data buffer in the descriptor or to the next descriptor.

Table 29-6 Receive Descriptor 3(RDES3)

Bit	Description
31:0	Buffer 2 Address Pointer (Next Descriptor Address) These bits indicate the physical address of Buffer 2 when a descriptor ring structure is used. If the Second Address Chained (RDES1[24]) bit is set, this address contains the pointer to the physical memory where the Next Descriptor is present. If RDES1[24] is set, the buffer (Next Descriptor) address pointer must be bus width-aligned (RDES3[3, 2, or 1:0] = 0, corresponding to a bus width of 128, 64, or 32. LSBs are ignored internally.) However, when RDES1[24] is reset, there are no limitations on the RDES3 value, except for the following condition: The DMA uses the configured value for its buffer address generation when the RDES3 value is used to store the start of frame. The DMA ignores RDES3[3, 2, or 1:0] (corresponding to a bus width of 128, 64, or 32) if the address pointer is to a buffer where the middle or last part of the frame is stored.

29.3.2 Transmit Descriptor

The descriptor addresses must be aligned to the bus width used (32/64/128). **Figure 29-7** shows the transmit descriptor format in little-endian mode with a 32-bit data bus.

Each descriptor is provided with two buffers, two byte-count buffers, and two address pointers, which enable the adapter port to be compatible with various types of memory-management schemes.

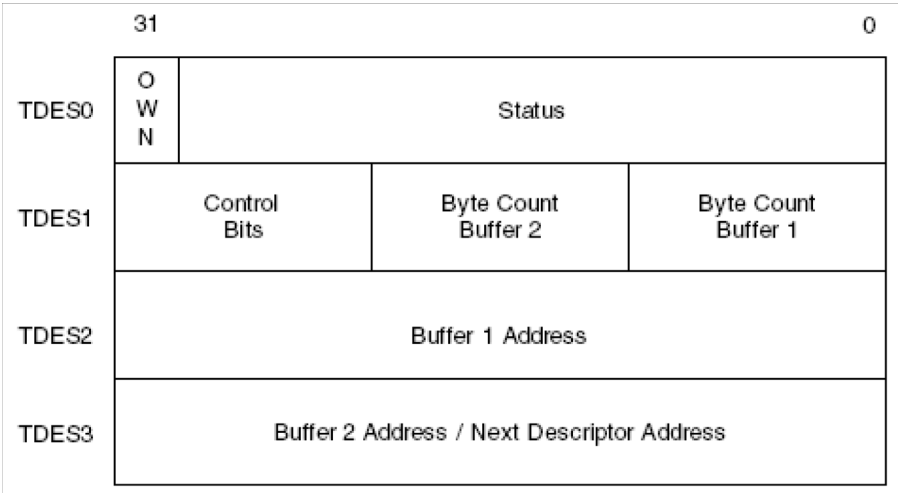


Figure 29-7 Transmit Descriptor Format

29.3.2.1 Transmit Descriptors 0(TDES0)

TDES0 contains the transmitted frame status and the descriptor ownership information.

Table 29-7 Transmit Descriptor 0

Bit	Description
31	OWN: Own Bit When set, this bit indicates that the descriptor is owned by the DMA. When this bit is reset, this bit indicates that the descriptor is owned by the Host. The DMA clears this bit either when it completes the frame transmission or when the buffers allocated in the descriptor are empty. The ownership bit of the First Descriptor of the frame should be set after all subsequent descriptors belonging to the same frame have been set. This avoids a possible race condition between fetching a descriptor and the driver setting an ownership bit.
30:18	Reserved
17	TTSS: Tx Timestamp Status This status bit indicates that a timestamp has been captured for the corresponding transmit frame. When this bit is set, TDES2 and TDES3 have timestamp values that were captured for the transmit frame. This field is valid only when the Last Segment control bit (TDES1[30]) in a descriptor is set. This bit is valid only when IEEE1588 time stamping feature is enabled; otherwise, it is reserved.
16	IHE: IP Header Error When set, this bit indicates that the Checksum Offload engine detected an IP header error and consequently did not modify the transmitted frame for any checksum insertion. This bit is valid only when Full Checksum Offload is enabled; otherwise, it is reserved.
15	ES: Error Summary Indicates the logical OR of the following bits: TDES0[14]: Jabber Timeout TDES0[13]: Frame Flush TDES0[11]: Loss of Carrier TDES0[10]: No Carrier TDES0[9]: Late Collision TDES0[8]: Excessive Collision TDES0[2]: Excessive Deferral TDES0[1]: Underflow Error
14	JT: Jabber Timeout When set, this bit indicates the MAC transmitter has experienced a jabber time-out. This bit is only set when the MAC configuration register's JD bit is not set.
13	FF: Frame Flushed When set, this bit indicates that the DMA/MTL flushed the frame due to a SW flush command given by the CPU.

Table 29-7 Transmit Descriptor 0

Bit	Description
12	PCE: Payload Checksum Error This bit, when set, indicates that the Checksum Offload engine had a failure and did not insert any checksum into the encapsulated TCP, UDP, or ICMP payload. This failure can be either due to insufficient bytes, as indicated by the IP Header's Payload Length field, or the MTL starting to forward the frame to the MAC transmitter in Store-and-Forward mode without the checksum having been calculated yet. This second error condition only occurs when the Transmit FIFO depth is less than the length of the Ethernet frame being transmitted: to avoid deadlock, the MTL starts forwarding the frame when the FIFO is full, even in Store-and-Forward mode. When the Full Checksum Offload engine is not enabled during configuration, this bit is reserved.
11	LC: Loss of Carrier When set, this bit indicates that Loss of Carrier occurred during frame transmission (that is, the gmii_crs_i signal was inactive for one or more transmit clock periods during frame transmission). This is valid only for the frames transmitted without collision and when the MAC operates in Half-Duplex Mode.
10	NC: No Carrier When set, this bit indicates that the carrier sense signal from the PHY was not asserted during transmission.
9	LC: Late Collision When set, this bit indicates that frame transmission was aborted due to a collision occurring after the collision window (64 byte times including Preamble in MII Mode and 512 byte times including Preamble and Carrier Extension in MACMII Mode). Not valid if Underflow Error is set.
8	EC: Excessive Collision When set, this bit indicates that the transmission was aborted after 16 successive collisions while attempting to transmit the current frame. If the DR (Disable Retry) bit in the MAC Configuration Register is set, this bit is set after the first collision and the transmission of the frame is aborted.
7	VF: VLAN Frame When set, this bit indicates that the transmitted frame was a VLAN-type frame.
6:3	CC: Collision Count T This 4-bit counter value indicates the number of collisions occurring before the frame was transmitted. The count is not valid when the Excessive Collisions bit (TDES0[8]) is set.
2	ED: Excessive Deferral When set, this bit indicates that the transmission has ended because of excessive deferral of over 24,288 bit times (155,680 bits times in 1000-Mbps mode, or in Jumbo Frame enabled mode) if the Deferral Check (DC) bit is set high in the MAC Control Register.
1	UF: Underflow Error When set, this bit indicates that the MAC aborted the frame because data arrived late from the Host memory. Underflow Error indicates that the DMA encountered an empty Transmit Buffer while transmitting the frame. The transmission process enters the suspended state and sets both Transmit Underflow (Register 5[5]) and Transmit Interrupt (Register 5[0]).
0	DB: Deferred Bit When set, this bit indicates that the MAC defers before transmission because of the presence of carrier. This bit is valid only in Half-Duplex mode.

29.3.2.2 Transmit Descriptors 1(TDES1)

TDES1 contains the buffer sizes and other bits which control the descriptor chain/ring and the frame being transferred.

Table 29-8 Transmit Descriptor 1

Bit	Description
31	IC: Interrupt on Completion When set, this bit sets Transmit Interrupt (Register 5[0]) after the present frame has been transmitted.
30	LS: Last Segment When set, this bit indicates that the buffer contains the last segment of the frame. When this bit is set, TBS1: Transmit Buffer 1 Size or TBS2: Transmit Buffer 2 Size field should have a non-zero value.
29	FS: First Segment When set, this bit indicates that the buffer contains the first segment of a frame.
28:27	CIC: Checksum Insertion Control These bits control the insertion of checksums in Ethernet frames that encapsulate TCP, UDP, or ICMP over IPv4 or IPv6 as described below. 2'b00: Do nothing. Checksum Engine is bypassed 2'b01: Insert IPv4 header checksum. Use this value to insert IPv4 header checksum when the frame encapsulates an IPv4 datagram. 2'b10: Insert TCP/UDP/ICMP checksum. The checksum is calculated over the TCP, UDP, or ICMP segment only and the TCP, UDP, or ICMP pseudo-header checksum is assumed to be present in the corresponding input frame's Checksum field. An IPv4 header checksum is also inserted if the encapsulated datagram conforms to IPv4. 2'b11: Insert a TCP/UDP/ICMP checksum that is fully calculated in this engine. In other words, the TCP, UDP, or ICMP pseudo-header is included in the checksum calculation, and the input frame's corresponding Checksum field has an all-zero value. An IPv4 Header checksum is also inserted if the encapsulated datagram conforms to IPv4. The Checksum engine detects whether the TCP, UDP, or ICMP segment is encapsulated in IPv4 or IPv6 and processes its data accordingly.
26	DC: Disable CRC When set, the MAC does not append the Cyclic Redundancy Check (CRC) to the end of the transmitted frame. This is valid only when the first segment (TDES1[29]).
25	TER: Transmit End of Ring When set, this bit indicates that the descriptor list reached its final descriptor. The returns to the base address of the list, creating a descriptor ring.
24	TCH: Second Address Chained When set, this bit indicates that the second address in the descriptor is the Next Descriptor address rather than the second buffer address. When TDES1[24] is set, TBS2 (TDES1[21–11]) are “don't care” values. TDES1[25] takes precedence over TDES1[24].

Table 29-8 Transmit Descriptor 1

Bit	Description
23	DP: Disable Padding When set, the MAC does not automatically add padding to a frame shorter than 64 bytes. When this bit is reset, the DMA automatically adds padding and CRC to a frame shorter than 64 bytes and the CRC field is added despite the state of the DC (TDES1[26]) bit. This is valid only when the first segment (TDES1[29]) is set.
22	TTSE: Transmit Timestamp Enable When set, this bit enables IEEE1588 hardware time stamping for the transmit frame referenced by the descriptor. This field is valid only when the First Segment control bit (TDES1[29]) is set.
21:11	TBS2: Transmit Buffer 2 Size These bits indicate the Second Data Buffer in bytes. This field is not valid if TDES1[24] is set.
10:0	TBS1: Transmit Buffer 1 Size These bits indicate the First Data Buffer byte size. If this field is 0, the DMA ignores this buffer and uses Buffer 2 or next descriptor depending on the value of TCH (Bit 24).

29.3.2.3 Transmit Descriptors 2(TDES2)

TDES2 contains the address pointer to the first buffer of the descriptor.

Table 29-9 Transmit Descriptor 2

Bit	Description
31:0	Buffer 1 Address Pointer These bits indicate the physical address of Buffer 1. There is no limitation on the buffer address alignment.

29.3.2.4 Transmit Descriptors 3(TDES3)

Table 29-10 Transmit Descriptor 3

Bit	Description
31:0	Buffer 2 Address Pointer (Next Descriptor Address) Indicates the physical address of Buffer 2 when a descriptor ring structure is used. If the Second Address Chained (TDES1[24]) bit is set, this address contains the pointer to the physical memory where the Next Descriptor is present. The buffer address pointer must be aligned to the bus width only when TDES1[24] is set. (LSBs are ignored internally.)

29.4 Continuous receive data mode

A continuous receive data mode (CRM) is developed to make rx data be stored continuously in rx data buffer.

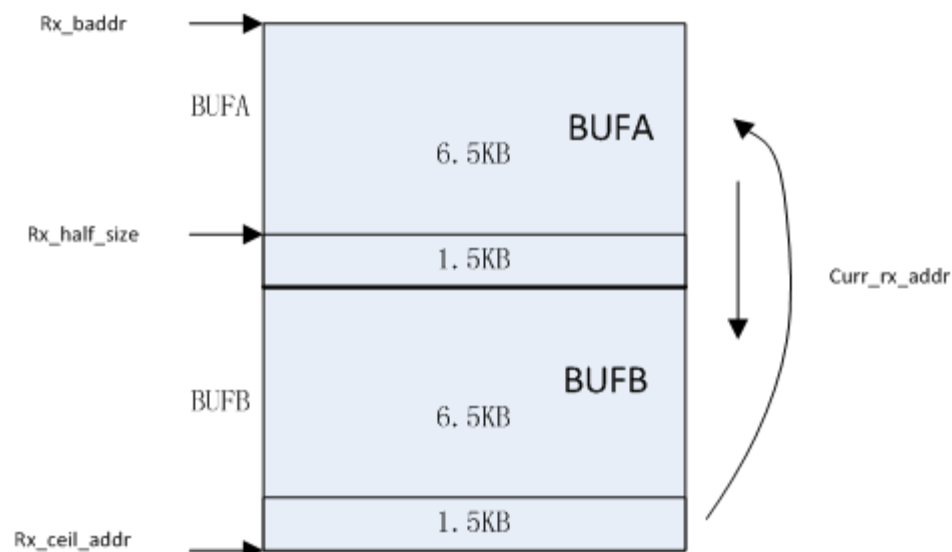


Figure 29-8 CRM 2-pingpong-buffer

The figure above shows the 2-pingpong-buffer located in system memory. The total rx buffer is divided into 2 single buffers which are buffer A and buffer B. The flow of the 2-pingpong-buffer is: DMA stores rx data in buffer A continuously (word aligned without interval). There is a margin size (typically max MAC frame size) for each buffer to avoid frame crossing buffers. When all rx data in buffer A exceeds rx_half_size (such as 8KB - 1.5KB = 6.5KB), DMA continues to store current frame but won't exceed buffer B base address, and switches base write address to buffer B base address for next frame if buffer B is available. After buffer A is full, the first address and the last address of all rx data in buffer A can be read from registers. Software clears buffer A full status to make buffer A available again. When all rx data in buffer B exceeds rx_half_size (such as 8KB - 1.5KB = 6.5KB), DMA continues to store current frame but won't exceed buffer ceil address, and switches base write address to buffer A base address for next frame if buffer A is available. After buffer B is full, the first address and the last address of all rx data in buffer B can be read from registers. Software clears buffer B full status to make buffer B available again.

In this mode rx descriptors, in which the addresses of RDES1 and RDES2 are exchanged, must work in "second address chained" mode. And the buff_size should be configured with max frame size which is typically 1.5KB. After each frame is stored into rx data buffer, DMA writes frame_length of current frame to the first word RDES0, and current frame base address "current_frame_baddr" to the second word RDES1. Software can read these rx descriptors to process rx data frames stored in rx buffer.

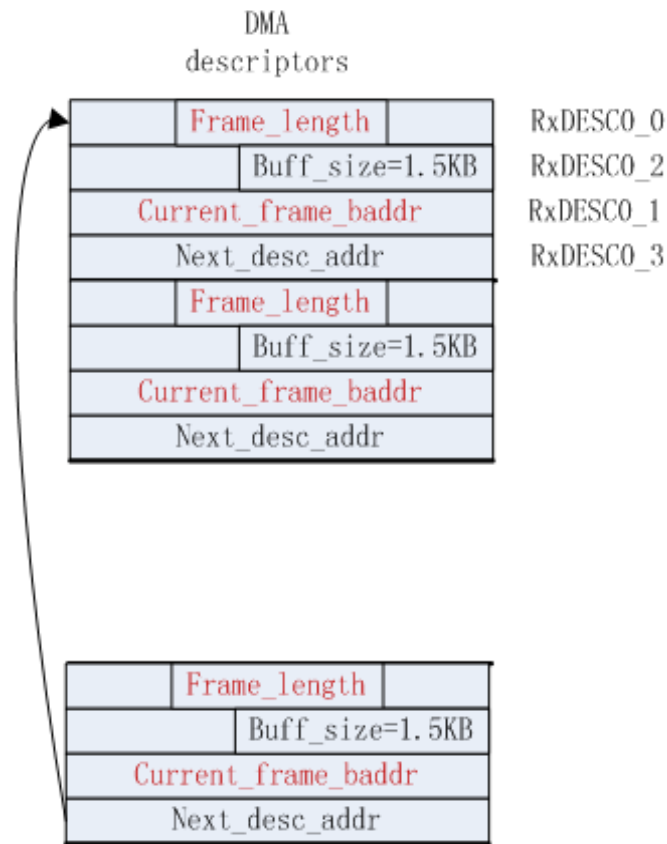


Figure 29-9 CRM descriptors

With this mode, software is easier to process too many small-sized frames using less rx descriptors. After all config is done, software waits for buffer A full, then read the first and last descriptor address for rx data in buffer A to process data in buffer A, and reinitializes those descriptors used. Then software waits for buffer B full, then read the first and last descriptor address for rx data in buffer A to process data in buffer B, and reinitializes those descriptors used. If all two buffers are full and software doesn't process the data in time, DMA goes into rx suspend mode (while rx desc unavailable status may not appear). To restart the rx DMA, software should write RPDR register. DMA continues to store rx data into buffer A or buffer B according to current buffer being written.

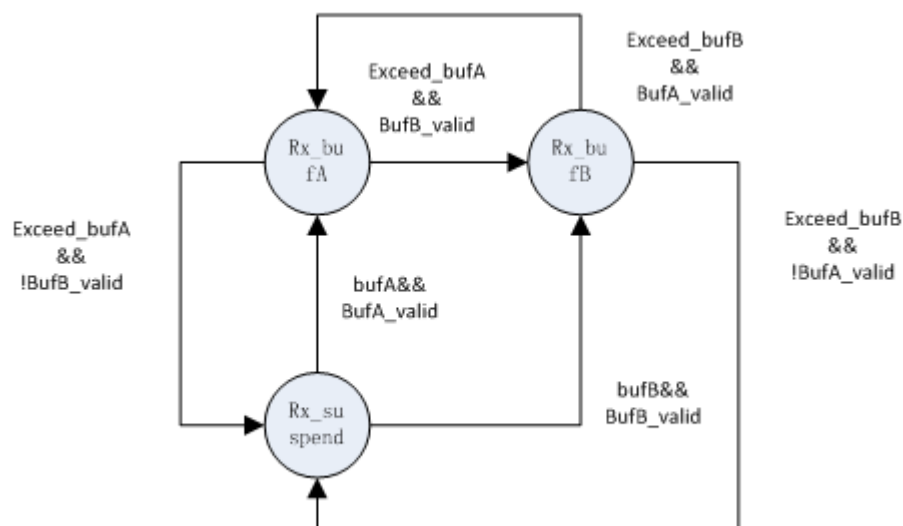


Figure 29-10 CRM 2-pingpong-buffer states

29.5 Register

29.5.1 Register Map

The register maps in this section provide high-level summaries of each register or group of registers. The register names in the register maps (tables) are cross-referenced to the detailed register descriptions in **29.5**. To view the detailed description of a register, double-click the register name.

29.5.1.1 DMA Register Map

Table 29-11 provides the address map of the DMA registers

Table 29-11 DMA Register Map

Register No	Offset Address	Register Name and Description
0	0x1000	Register 0 (Bus Mode Register) Controls the Host Interface Mode.
1	0x1004	Register 1 (Transmit Poll Demand Register) Used by the host to instruct the DMA to poll the Transmit Descriptor list.

Table 29-11 DMA Register Map

Register No	Offset Address	Register Name and Description
2	0x1008	Register 2 (Receive Poll Demand Register) Used by the Host to instruct the DMA to poll the Receive Descriptor list.
3	0x100C	Register 3 (Receive Descriptor List Address Register) Points the DMA to the start of the Receive Descriptor list.
4	0x1010	Register 4 (Transmit Descriptor List Address Register) Points the DMA to the start of the Transmit Descriptor list.
5	0x1014	Register 5 (Status Register) The Software driver (application) reads this register during interrupt service routine or polling to determine the status of the DMA.
6	0x1018	Register 5 (Status Register) The Software driver (application) reads this register during interrupt service routine or polling to determine the status of the DMA.
7	0x101C	Register 7 (Interrupt Enable Register) Enables the interrupts reported by the Status Register.
8	0x1020	Register 8 (Missed Frame and Buffer Overflow Counter Register) Contains the counters for discarded frames because no host Receive Descriptor was available, and discarded frames because of Receive FIFO Overflow.
9	0x1024	Register 9 (Receive Interrupt Watchdog Timer Register) Watchdog time-out for Receive Interrupt (RI) from DMA.
10	0x1028	Reserved
11	0x102C	Reserved
12-17	0x1030-0x1044	Reserved
18	0x1048	Register 18 (Current Host Transmit Descriptor Register) Points to the start of current Transmit Descriptor read by the DMA.
19	0x104C	Register 19 (Current Host Receive Descriptor Register) Points to the start of current Receive Descriptor read by the DMA.
20	0x1050	Register 20 (Current Host Transmit Buffer Address Register) Points to the current Transmit Buffer address read by the DMA.
21	0x1054	Register 21 (Current Host Receive Buffer Address Register) Points to the current Receive Buffer address read by the DMA.
22	0x1058	Register 22 (HW Feature Register) Indicates the presence of the optional features of the core.
23	0x105c	Register 23 (continuous receive mode base address register) A base address configured for continuous receive mode.
24	0x1060	Register 24 (continuous rx mode register) continuous rx data mode enable.
25	0x1064	Register 25 (continuous rx mode current address register) Current rx data dma write address.
26	0x1068	Register 26 (continuous rx mode ceil address register) max address for 2-pingpong-buffer in continuous rx mode.
27	0x106c	Register 27 (continuous rx mode half size register) size of single buffer of 2-pingpong-buffer.

Table 29-11 DMA Register Map

Register No	Offset Address	Register Name and Description
28	0x1070	Register 28 (continuous rx mode buffer A descriptor start address register) The start desc base address of rx data in buffer A
29	0x1074	Register 29 (continuous rx mode buffer B descriptor start address register) The start desc base address of rx data in buffer B
30	0x1078	Register 30 (continuous rx mode buffer A descriptor end address register) The end desc base address of rx data in buffer A
31	0x107c	Register 31 (continuous rx mode buffer B descriptor end address register) The end desc base address of rx data in buffer B
Others	Others	Reserved

29.5.1.2 MAC Register Map

Table 29-12 provides the address map of the MAC core registers.

Table 29-12 MAC Register Map

Register No	Offset Address	Register Name and Description
0	0x0000	Register 0 (MAC Configuration Register) This is the operation mode register for the MAC.
1	0x0004	Register 1 (MAC Frame Filter) Contains the frame filtering controls.
2	0x0008	Register 2 (Hash Table High Register) Contains the higher 32 bits of the Multicast Hash table.
3	0x000C	Register 3 (Hash Table Low Register) Contains the lower 32 bits of the Multicast Hash table.
4	0x0010	Register 4 (MACMII Address Register) Controls the management cycles to an external PHY.
5	0x0014	Register 5 (MACMII Data Register) Contains the data to be written to or read from the PHY register.
6	0x0018	Register 6 (Flow Control Register) Controls the generation of control frames.
7	0x001C	Register 7 (VLAN Tag Register) Identifies IEEE 802.1Q VLAN type frames.
8	0x0020	Register 8 (Version Register) Identifies the version of the Core.
9	0x0024	Register 9 (Debug Register) MAC gives the status of various internal blocks for debugging.

Table 29-12 MAC Register Map

Register No	Offset Address	Register Name and Description
10	0x0028	Remote Wake-Up Frame Filter Register This is the address through which the application writes or reads the remote wake-up frame filter registers (wkupfilter_reg). The wkupfilter_reg register is a pointer to eight wkupfilter_reg registers. The wkupfilter_reg register is loaded by sequentially loading the eight register values. Eight sequential writes to this address (0x0028) writes all wkupfilter_reg registers. Similarly, eight sequential reads from this address (0x0028) read all wkupfilter_reg registers. This register contains the higher 16 bits of the seventh MAC address.
11	0x002C	PMT Control and Status Register
12	0x0030	Reserved
13	0x0034	Reserved
14	0x0038	Register 14 (Interrupt Status Register) Contains the interrupt status.
15	0x003C	Register 15 (Interrupt Mask Register) Contains the masks for generating the interrupts.
16	0x0040	Register 16 (MAC Address0 High Register) Contains the higher 16 bits of the first MAC address.
17	0x0044	Register 17 (MAC Address0 Low Register) Contains the lower 32 bits of the first MAC address.
18	0x0048	Register 18 (MAC Address1 High Register) Contains the higher 16 bits of the second MAC address.
19	0x004C	Register 19 (MAC Address1 Low Register) Contains the lower 32 bits of the second MAC address.
20	0x0050	MAC Address2 High Register Contains the lower 32 bits of the third MAC address.
21	0x0054	MAC Address2 Low Register Contains the lower 32 bits of the third MAC address.
22	0x0058	MAC Address3 High Register Contains the higher 16 bits of the fourth MAC address.
23	0x005C	MAC Address3 Low Register Contains the lower 32 bits of the fourth MAC address.
24	0x0060	MAC Address4 High Register Contains the higher 16 bits of the fifth MAC address.
25	0x0064	MAC Address4 Low Register Contains the lower 32 bits of the fifth MAC address.
26	0x0068	MAC Address5 High Register Contains the higher 16 bits of the sixth MAC address.
27	0x006C	MAC Address5 Low Register Contains the lower 32 bits of the sixth MAC address.
28	0x0070	MAC Address6 High Register Contains the higher 16 bits of the seventh MAC address.

Table 29-12 MAC Register Map

Register No	Offset Address	Register Name and Description
29	0x0074	MAC Address6 Low Register Contains the lower 32 bits of the seventh MAC address.
30	0x0078	MAC Address6 Low Register Contains the lower 32 bits of the seventh MAC address.
31	0x007C	MAC Address7 Low Register Contains the lower 32 bits of the eighth MAC address.
32	0x0080	MAC Address8 High Register Contains the higher 16 bits of the ninth MAC address.
33	0x0084	MAC Address8 Low Register Contains the lower 32 bits of the ninth MAC address.
34	0x0088	MAC Address9 High Register Contains the higher 16 bits of the tenth MAC address.
35	0x008C	MAC Address9 Low Register Contains the lower 32 bits of the tenth MAC address.
36	0x0090	MAC Address10 High Register Contains the higher 16 bits of the eleventh MAC address.
37	0x0094	MAC Address10 Low Register Contains the lower 32 bits of the eleventh MAC address. .
38	0x0098	MAC Address11 High Register This register contains the higher 16 bits of the twelfth MAC address.
39	0x009C	MAC Address11 Low Register Contains the lower 32 bits of the twelfth MAC address.
40	0x00A0	MAC Address12 High Register Contains the higher 16 bits of the thirteenth MAC address.
41	0x00A4	MAC Address12 Low Register Contains the lower 32 bits of the thirteenth MAC address.
42	0x00A8	MAC Address13 High Register Contains the higher 16 bits of the fourteenth MAC address.
43	0x00AC	MAC Address13 High Register Contains the higher 16 bits of the fourteenth MAC address.
44	0x00B0	MAC Address14 High Register Contains the higher 16 bits of the fifteenth MAC address.
45	0x00B4	MAC Address14 Low Register Contains the lower 32 bits of the fifteenth MAC address.
46	0x00B8	MAC Address15 High Register Contains the higher 16 bits of the sixteenth MAC address.
47	0x00BC	MAC Address15 Low Register Contains the lower 32 bits of the sixteenth MAC address.
48	0x0800	MAC Address 16 High Register Contains the higher 16 bits of the seventeenth MAC address.
49	0x0804	MAC Address 16 Low Register Contains the lower 32 bits of the seventeenth MAC address.

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Table 29-12 MAC Register Map

Register No	Offset Address	Register Name and Description
50	0x0808	MAC Address 17 High Register Contains the higher 16 bits of the eighteenth MAC address.
51~78	0x080C~0x0878	MAC Address 17~MAC Address 31
79	0x087C	MAC Address 31 Low Register Contains the lower 32 bits of the thirty-second MAC address.
Others	Others	Reserved

29.5.1.3 MMC Register Map

Table 29-13 MMC Register Map

Register No	Address Offset	Register Name	Register Name and Description
0	0x011C	txbroadcastframes_g	Number of good broadcast frames transmitted.
1	0x0120	txmulticastframes_g	Number of good multicast frames transmitted.
2	0x0148	txunderflowerror	Number of frames aborted because of frame underflow error.
3	0x014C	txsinglecollision_g	Number of successfully transmitted frames after a single collision in Half-duplex mode.
4	0x0150	txmulticollision_g	Number of successfully transmitted frames after more than a single collision in Half-duplex mode.
5	0x0154	txdeferred	Number of successfully transmitted frames after a deferral in Half-duplex mode.
6	0x0158	txlatecollision	Number of frames aborted because of late collision error.
7	0x015C	txexcessivecollision	Number of frames aborted because of excessive (16) collision errors.
8	0x0170	txpauseframes	Number of good PAUSE frames transmitted.
9	0x018C	rxbroadcastframes_g	Number of good broadcast frames received.
10	0x0190	rxmulticastframes_g	Number of good multicast frames received.

Table 29-13 MMC Register Map

Register No	Address Offset	Register Name	Register Name and Description
11	0x019C	rxrunterror	Number of frames received with runt (<64 bytes and CRC error) error.
12	0x01A0	rxjabbererror	Number of giant frames received with length (including CRC) greater than 1,518 bytes (1,522 bytes for VLAN tagged) and with CRC error. If Jumbo Frame mode is enabled, then frames of length greater than 9,018 bytes (9,022 for VLAN tagged) are considered as giant frames.
13	0x01A4	rxundersize_g	Number of frames received with length less than 64 bytes, without any errors.
14	0x01A8	rxoversize_g	Number of frames received with length greater than the maxsize (1,518 or 1,522 for VLAN tagged frames), without errors.
15	0x01C4	rxunicastframes_g	Number of good unicast frames received.
16	0x01D0	rxpauseframes	Number of good and valid PAUSE frames received.
Others	Others	Others	Reserved

29.5.2 Register Descriptions

The Access column of each register description that follows specifies how the application and the core can access the register fields of the CSRs.

The Access column uses the following conventions:

Read Only (RO)

Register field can only be read by the application. Writes to read-only fields have no effect.

Read and Write (R_W)

Register field can be read and written by the application. The application can set this field by writing 1'b1 and can clear it by writing 1'b0.

Read Only and Write Trigger (RO_WT)

Register field can be read by the application, and when a write operation is performed with any data value, an event is triggered, as explained in the field's description. (Example: Tx Poll Demand register)

Read, Write Set, and Self Clear (R_WS_SC)

Register field can be read by the application (Read), can be set to 1'b1 by the application with a register write of 1'b1 (Write Set), and is cleared to 1'b0 by the core (Self Clear). The application cannot clear this type of field, and a register write of 1'b0 to this bit has no effect on this field. The conditions under which the core clears this field are explained in detail in the field's description. (Example, soft-reset signals)

29.5.2.1 DMA Register Description

This section defines the bits for each DMA register. The write data inputs to the DMA registers are qualified with the corresponding mci_be_i signal inputs (MCI interface). Thus, non-32 bit accesses are allowed as long as the address is Word-aligned. For the APB interface, only 32-bit accesses are possible, while for AHB slave interfaces, byte, half-word, or word accesses are possible.

29.5.2.1.1 Register 0(Bus Mode Register)

The Bus Mode register establishes the bus operating modes for the DMA.

Table 29-14 Register 0(Bus Mode Register)

Field	Description	Reset	Access
31:30	Reserved	0	RO
29:28	PRWMAC: Channel Priority Weights Sets the priority weights for channel 0 during the round-robin arbitration between the DMA channels for the system bus. Value Priority Weight 00 1 01 2 10 3 11 4		R_W
27	This bit is reserved and is read-only (RO).	0	RO
26	reserved	0	RO
25	AAL: Address-Aligned Beats When this bit is set high and the FB bit equals 1, the AHB/AXI interface generates all bursts aligned to the start address LS bits. If the FB bit is equal to 0, the first burst (accessing the data buffer's start address) is not aligned, but subsequent bursts are aligned to the address.	0	R_W
24	8xPBL Mode When set high, this bit multiplies the PBL value programmed (bits [22:17] and bits [13:8]) eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value.	0	R_W

Table 29-14 Register 0(Bus Mode Register)

Field	Description	Reset	Access
23	USP: Use Separate PBL When set high, this bit configures the RxDMA to use the value configured in bits [22:17] as PBL while the PBL value in bits [13:8] is applicable to TxDMA operations only. When reset to low, the PBL value in bits [13:8] is applicable for both DMA engines.	0	R_W
22:17	RPBL: RxDMA PBL These bits indicate the maximum number of beats to be transferred in one RxDMA transaction. This is the maximum value that is used in a single block Read/Write. The RxDMA always attempts to burst as specified in RPBL every time it starts a Burst transfer on the host bus. RPBL can be programmed with permissible values of 1, 2, 4, 8, 16, and 32. Any other value results in undefined behavior. These bits are valid and applicable only when USP is set high.	01H	R_W
16	FB: Fixed Burst This bit controls whether the AXI Master interface performs fixed burst transfers or not. When reset, the AXI uses SINMACLE and INCR burst transfer operations.	0	R_W
15:14	reserved	0	RO

Table 29-14 Register 0(Bus Mode Register)

Field	Description	Reset	Access								
13:8	PBL: Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA transaction. This is the maximum value that is used in a single block Read/Write. The DMA always attempts to burst as specified in PBL each time it starts a Burst transfer on the host bus. PBL can be programmed with permissible values of 1, 2, 4, 8, 16, and 32. Any other value results in undefined behavior. When USP is set high, this PBL value is applicable for TxDMA transactions only. If the number of beats to be transferred is more than 32, then perform the following steps: 1. Set the 8xPBL mode. 2. Set the PBL. For example, if the maximum number of beats to be transferred is 64, then first set 8xPBL to 1 and then set PBL to 8. The PBL values have the following limitation: The maximum number of possible beats (PBL) is limited by the size of the Tx FIFO and Rx FIFO in the MTL layer and the data bus width on the DMA. The FIFO has a constraint that the maximum beat supported is half the depth of the FIFO, except when specified. For different data bus widths and FIFO sizes, the valid PBL range (including x8 mode) is provided in the following table. If the PBL is common for both transmit and receive DMA, the minimum Rx FIFO and Tx FIFO depths must be considered. <table> <tr> <th>Data Bus Width</th><th>FIFO Depth</th><th>Valid PBL Range in Half Duplex Mode</th><th>Valid PBL Range in Half Duplex Mode</th></tr> <tr> <td>64</td><td>4 KB</td><td>All</td><td>All</td></tr> </table>	Data Bus Width	FIFO Depth	Valid PBL Range in Half Duplex Mode	Valid PBL Range in Half Duplex Mode	64	4 KB	All	All	0	R_W
Data Bus Width	FIFO Depth	Valid PBL Range in Half Duplex Mode	Valid PBL Range in Half Duplex Mode								
64	4 KB	All	All								
7	ATDS: Alternate Descriptor Size When set, the size of the alternate descriptor increases to 32 bytes (8 DWORDS). This is required when the Advanced Timestamp feature or Full IPC Offload Engine is enabled in the receiver. This bit is present only when you select the Alternate Descriptor feature and any one of the following features during configuration: • Advanced Timestamp feature • IPC Full Checksum Offload (type 2) feature Otherwise, this bit is reserved and is read-only. When reset, the descriptor size reverts back to 4 DWORDs (16 bytes). This bit preserves the backward compatibility for the descriptor size. In versions prior to 3.50a, the descriptor size is 16 bytes for both normal and enhanced descriptor. In 3.50a, descriptor size is increased to 32 bytes because of the Advanced Timestamp and IPC Full Checksum Offload features. The enhanced descriptor is not required if the Advanced Timestamp and IPC Full Checksum Offload features are not enabled. In such case, you can use the 16 bytes descriptor to save 4 bytes of memory.	0	R_W								

Table 29-14 Register 0(Bus Mode Register)

Field	Description	Reset	Access
6:2	DSL: Descriptor Skip Length This bit specifies the number of Word/Dword/Lword (depending on 32/64/128-bit bus) to skip between two unchained descriptors. The address skipping starts from the end of current descriptor to the start of next descriptor. When DSL value is equal to zero, then the descriptor table is taken as contiguous by the DMA, in Ring mode.	00H	R_W
1	reserved	0	RO
0	SWR: Software Reset When this bit is set, the MAC DMA Controller resets all MAC Subsystem internal registers and logic. It is cleared automatically after the reset operation has completed in all of the core clock domains. Read a 0 value in this bit before re-programming any register of the core. Note: The reset operation is completed only when all the resets in all the active clock domains are de-asserted. Therefore, it is essential that all the PHY inputs clocks (applicable for the selected PHY interface) are present for software reset completion.	0	R_WS_SC

29.5.2.1.2 Resister 1(Transmit Poll Demand Register)

The Transmit Poll Demand register enables the Transmit DMA to check whether or not the current descriptor is owned by DMA. The Transmit Poll Demand command is given to wake up the TxDMA if it is in Suspend mode. The TxDMA can go into Suspend mode because of an Underflow error in a transmitted frame or because of the unavailability of descriptors owned by Transmit DMA. You can give this command anytime and the TxDMA resets this command once it starts re-fetching the current descriptor from host memory.

Table 29-15 Register 1(Transmit Poll Demand Register)

Field	Description	Reset	Access
31:0	TPD: Transmit Poll Demand When these bits are written with any value, the DMA reads the current descriptor pointed to by Register 18. If that descriptor is not available (owned by Host), transmission returns to the Suspend state and DMA Register 5[2] is asserted. If the descriptor is available, transmission resumes.	0000_0000 H	RO_WT

29.5.2.1.3 Resister 2(Receive Poll Demand Register)

The Receive Poll Demand register enables the receive DMA to check for new descriptors. This command is given to wake up the RxDMA from SUSPEND state. The RxDMA can go into SUSPEND state only because of the unavailability of descriptors owned by it.

Table 29-16 Register 1(Receive Poll Demand Register)

Field	Description	Reset	Access
31:0	RPD: Receive Poll Demand When these bits are written with any value, the DMA reads the current descriptor pointed to by Register 19. If that descriptor is not available (owned by Host), reception returns to the Suspended state and Register 5[7] is not asserted. If the descriptor is available, the Receive DMA returns to the active state.	0000_0000 H	RO_WT

29.5.2.1.4 Register 3(Receive Descriptor List Address Register)

The Receive Descriptor List Address register points to the start of the Receive Descriptor List. The descriptor lists reside in the host's physical memory space and must be Word/Dword/Lword-aligned (for 32/64/128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to Register 3 is permitted only when reception is stopped. When stopped, Register 3 must be written to before the receive Start command is given.

You can write to Register 3 only when Rx DMA has stopped, that is, bit 1 (SR) is set to zero in Register 6 **29.5.2.1.7**(Operation Mode Register). When stopped, Register 3 can be written with a new descriptor list address. When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.

If the Register 3 is not changed when the SR bit is set to 0, then the DMA takes the descriptor address where it was stopped earlier.

Table 29-17 Register 3(Receive Descriptor List Address Register)

Field	Description	Reset	Access
31:0	Start of Receive List This field contains the base address of the First Descriptor in the Receive Descriptor list. The LSB bits [1/2/3:0] for 32/64/128-bit bus width are ignored and are taken as all-zero by the DMA internally. Therefore, these LSB bits are Read- Only (RO).	0000_0000 H	R_W

29.5.2.1.5 Register 4(Transmit Descriptor List Address Register)

The Transmit Descriptor List Address register points to the start of the Transmit Descriptor List. The descriptor lists reside in the host's physical memory space and must be Word/DWORD/LWORD-aligned (for 32/64/128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.

You can write to Register 4 only when Tx DMA has stopped, that is, bit 13 (ST) is set to zero in Register 6 **29.5.2.1.7**(Operation Mode Register). When stopped, Register 4 can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly programmed descriptor base address.

If the Register 4 is not changed when the ST bit is set to 0, then the DMA takes the descriptor address where it was stopped earlier.

Table 29-18 Register 4(Transmit Descriptor List Address Register)

Field	Description	Reset	Access
31:0	Start of Receive List This field contains the base address of the First Descriptor in the Receive Descriptor list. The LSB bits [1/2/3:0] for 32/64/128-bit bus width are ignored and are taken as all-zero by the DMA internally. Therefore, these LSB bits are Read- Only (RO).	0000_0000 H	R_W

29.5.2.1.6 Register 5(Status Register)

The Status register contains all the status bits that the DMA reports to the host. The Software driver reads this register during an interrupt service routine or polling. Most of the fields in this register cause the host to be interrupted. Register 5 bits are not cleared when read. Writing 1'b1 to (unreserved) bits in Register 5[16:0] clears these bits and writing 1'b0 has no effect. Each field (bits[16:0]) can be masked by masking the appropriate bit in Register 7.

Table 29-19 Register 5(Status Register)

Field	Description	Reset	Access
31	CRMBBF: buffer B is full in continuous rx data mode	0	RO
30	CRMBAF: buffer A is full in continuous rx data mode	0	RO
29	Reserved	0	RO

Table 29-19 Register 5(Status Register)

Field	Description	Reset	Access
28	Reserved	0	RO
27	MACMI: MAC MMC Interrupt This bit reflects an interrupt event in the MMC module of the MAC core. The software must read the corresponding registers in the MAC core to get the exact cause of interrupt and clear the source of interrupt to make this bit as 1'b0. The interrupt signal from the MAC subsystem (sbd_intr_o) is high when this bit is high.	0	RO
26	Reserved	0	RO
25:23	EB: Error Bits These bits indicate the type of error that caused a Bus Error (e.g., error response on the AHB/AXI interface). These bits are valid only when bit 13 (FBI: Fatal Bus Error Interrupt) is set. This field does not generate an interrupt. Bit 23 1'b1 Error during data transfer by TxDMA 1'b0 Error during data transfer by RxDMA Bit 24 1'b1 Error during read transfer 1'b0 Error during write transfer Bit 25 1'b1 Error during descriptor access 1'b0 Error during data buffer access	0	RO
22:20	TS: Transmit Process State These bits indicate the Transmit DMA FSM state. This field does not generate an interrupt. 3'b000: Stopped; Reset or Stop Transmit Command issued. 3'b001: Running; Fetching Transmit Transfer Descriptor. 3'b010: Running; Waiting for status. 3'b011: Running; Reading Data from host memory buffer and queuing it to transmit buffer (Tx FIFO). 3'b100: TIME_STAMP write state. 3'b101: Reserved for future use. 3'b110: Suspended; Transmit Descriptor Unavailable or Transmit Buffer Underflow. 3'b111: Running; Closing Transmit Descriptor.	0	RO
19:17	RS: Receive Process State These bits indicate the Receive DMA FSM state. This field does not generate an interrupt. 3'b000: Stopped: Reset or Stop Receive Command issued. 3'b001: Running: Fetching Receive Transfer Descriptor. 3'b010: Reserved for future use. 3'b011: Running; Waiting for receive packet. 3'b100: Suspended: Receive Descriptor Unavailable. 3'b101: Running: Closing Receive Descriptor. 3'b110: TIME_STAMP write state. 3'b111: Running: Transferring the receive packet data from receive buffer to host memory.	0	RO

Table 29-19 Register 5(Status Register)

Field	Description	Reset	Access
16	NIS: Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in DMA Register 7: Register 5[0]: Transmit Interrupt Register 5[2]: Transmit Buffer Unavailable Register 5[6]: Receive Interrupt Register 5[14]: Early Receive Interrupt Only unmasked bits affect the Normal Interrupt Summary bit. This is a sticky bit and must be cleared (by writing a 1 to this bit) each time a corresponding bit that causes NIS to be set is cleared.	0	RO
15	AIS: Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in DMA Register 7: Register 5[1]: Transmit Process Stopped Register 5[3]: Transmit Jabber Timeout Register 5[4]: Receive FIFO Overflow Register 5[5]: Transmit Underflow Register 5[7]: Receive Buffer Unavailable Register 5[8]: Receive Process Stopped Register 5[9]: Receive Watchdog Timeout Register 5[10]: Early Transmit Interrupt Register 5[13]: Fatal Bus Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit and must be cleared each time a corresponding bit that causes AIS to be set is cleared.	0	R_SS_ WC
14	ERI: Early Receive Interrupt This bit indicates that the DMA had filled the first data buffer of the packet. Receive Interrupt Register 5[6] automatically clears this bit.	0	R_SS_ WC
13	FBI: Fatal Bus Error Interrupt This bit indicates that a bus error occurred, as detailed in [25:23]. When this bit is set, the corresponding DMA engine disables all its bus accesses.	0	R_SS_ WC
12:11	Reserved	0	R_SS_ WC
10	ETI: Early Transmit Interrupt This bit indicates that the frame to be transmitted was fully transferred to the MTL Transmit FIFO.	0	R_SS_ WC
9	RWT: Receive Watchdog Timeout This bit is asserted when a frame with a length greater than 2,048 bytes is received (10, 240 when Jumbo Frame mode is enabled).	0	R_SS_ WC
8	RPS: Receive Process Stopped This bit is asserted when the Receive Process enters the Stopped state.	0	R_SS_ WC

Table 29-19 Register 5(Status Register)

Field	Description	Reset	Access
7	RU: Receive Buffer Unavailable This bit indicates that the Next Descriptor in the Receive List is owned by the host and cannot be acquired by the DMA. Receive Process is suspended. To resume processing Receive descriptors, the host should change the ownership of the descriptor and issue a Receive Poll Demand command. If no Receive Poll Demand is issued, Receive Process resumes when the next recognized incoming frame is received. Register 5[7] is set only when the previous Receive Descriptor was owned by the DMA.	0	R_SS_ WC
6	RI: Receive Interrupt This bit indicates the completion of frame reception. Specific frame status information has been posted in the descriptor. Reception remains in the Running state.	0	R_SS_ WC
5	UNF: Transmit Underflow This bit indicates that the Transmit Buffer had an Underflow during frame transmission. Transmission is suspended and an Underflow Error TDES0[1] is set.	0	R_SS_ WC
4	OVF: Receive Overflow This bit indicates that the Receive Buffer had an Overflow during frame reception. If the partial frame is transferred to application, the overflow status is set in RDES0[11].	0	R_SS_ WC
3	TJT: Transmit Jabber Timeout This bit indicates that the Transmit Jabber Timer expired, meaning that the transmitter had been excessively active. The transmission process is aborted and placed in the Stopped state. This causes the Transmit Jabber Timeout TDES0[14] flag to assert.	0	R_SS_ WC
2	TU: Transmit Buffer Unavailable This bit indicates that the Next Descriptor in the Transmit List is owned by the host and cannot be acquired by the DMA. Transmission is suspended. Bits[22:20] explain the Transmit Process state transitions. To resume processing transmit descriptors, the host should change the ownership of the bit of the descriptor and then issue a Transmit Poll Demand command.	0	R_SS_ WC
1	TPS: Transmit Process Stopped This bit is set when the transmission is stopped.	0	R_SS_ WC
0	TI: Transmit Interrupt This bit indicates that frame transmission is finished and TDES1[31] is set in the First Descriptor.	0	R_SS_ WC

29.5.2.1.7 Register 6(Operation Mode Register)

The Operation Mode register establishes the Transmit and Receive operating modes and commands. Register 6 should be the last CSR to be written as part of DMA initialization. This register is also present in the MAC-MTL configuration with bits 24, 13, 2, and 1 unused and reserved.

Table 29-20 Register 6(Operation Mode Register)

Field	Description	Reset	Access
31:27	Reserved	000H	RO
26	DT: Disable Dropping of TCP/IP Checksum Error Frames When this bit is set, the core does not drop frames that only have errors detected by the Receive Checksum Offload engine. Such frames do not have any errors (including FCS error) in the Ethernet frame received by the MAC but have errors in the encapsulated payload only. When this bit is reset, all error frames are dropped if the FEF bit is reset. If the Full Checksum Offload engine (Type 2) is disabled, this bit is reserved (RO with value 1'b0).	0	R_W
25	RSF: Receive Store and Forward When this bit is set, the MTL only reads a frame from the Rx FIFO after the complete frame has been written to it, ignoring RTC bits. When this bit is reset, the Rx FIFO operates in Cut-Through mode, subject to the threshold specified by the RTC bits.	0	R_W
24	DFF: Disable Flushing of Received Frames When this bit is set, the RxDMA does not flush any frames because of the unavailability of receive descriptors/buffers as it does normally when this bit is reset.	0	R_W
23	reserved	0	RO
22	reserved	0	RO
21	TSF: Transmit Store and Forward When this bit is set, transmission starts when a full frame resides in the MTL Transmit FIFO. When this bit is set, the TTC values specified in Register 6[16:14] are ignored. This bit should be changed only when transmission is stopped.	0	R_W
20	FTF: Flush Transmit FIFO When this bit is set, the transmit FIFO controller logic is reset to its default values and thus all data in the Tx FIFO is lost/flushed. This bit is cleared internally when the flushing operation is completed fully. The Operation Mode register should not be written to until this bit is cleared. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt frame transmission. <i>Note:</i> The flush operation completes only after emptying the TxFIFO of its contents and all the pending Transmit Status of the transmitted frames are accepted by the host. In order to complete this flush operation, the PHY transmit clock (clk_tx_i) is required to be active.	0	R_WS_SC
19:17	Reserved	0	RO

Table 29-20 Register 6(Operation Mode Register)

Field	Description	Reset	Access
16:14	TTC: Transmit Threshold Control These three bits control the threshold level of the MTL Transmit FIFO. Transmission starts when the frame size within the MTL Transmit FIFO is larger than the threshold. In addition, full frames with a length less than the threshold are also transmitted. These bits are used only when the TSF bit (Bit 21) is reset. 000 64 001 128 010 192 011 256 100 40 101 32 110 24 111 16	0	R_W
13	ST: Start/Stop Transmission Command When this bit is set, transmission is placed in the Running state, and the DMA checks the Transmit List at the current position for a frame to be transmitted. Descriptor acquisition is attempted either from the current position in the list, which is the Transmit List Base Address set by Register 4, or from the position retained when transmission was stopped previously. If the current descriptor is not owned by the DMA, transmission enters the Suspended state and Transmit Buffer Unavailable (Register 5[2]) is set. The Start Transmission command is effective only when transmission is stopped. If the command is issued before setting DMA Register 4, then the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current frame. The Next Descriptor position in the Transmit List is saved, and becomes the current position when transmission is restarted. The stop transmission command is effective only when the transmission of the current frame is complete or the transmission is in the Suspended state.	0	R_W
12:11	RFD: Threshold for deactivating flow control (in both HD and FD) These bits control the threshold (Fill-level of Rx FIFO) at which the flow-control is de-asserted after activation. 00 Full minus 1 KB 01 Full minus 2 KB 10 Full minus 3 KB 11 Full minus 4 KB Note that the de-assertion is effective only after flow control is asserted.	0	R_W

Table 29-20 Register 6(Operation Mode Register)

Field	Description	Reset	Access
10:9	RFA: Threshold for activating flow control (in both HD and FD) These bits control the threshold (Fill level of Rx FIFO) at which flow control is activated. 00 Full minus 1 KB 01 Full minus 2 KB 10 Full minus 3 KB 11 Full minus 4 KB Note that the above only applies to Rx FIFOs of 4 KB or more when the EFC bit is set high.	0	R_W
8	EFC: Enable HW flow control When this bit is set, the flow control signal operation based on fill-level of Rx FIFO is enabled. When reset, the flow control operation is disabled. This bit is not used (reserved and always reset) when the Rx FIFO is less than 4 KB.	0	R_W
7	FEF: Forward Error Frames When this bit is reset, the Rx FIFO drops frames with error status (CRC error, collision error, MACMII_ER, giant frame, watchdog timeout, overflow). However, if the frame's start byte (write) pointer is already transferred to the read controller side (in Threshold mode), then the frames are not dropped. When FEF bit is set, all frames except runt error frames are forwarded to the DMA. But when Rx FIFO overflows when a partial frame is written, then such frames are dropped even when FEF is set.	0	R_W
6	FUF: Forward Undersized MACood Frames When set, the Rx FIFO forwards Undersized frames (frames with no Error and length less than 64 bytes) including pad-bytes and CRC). When reset, the Rx FIFO drops all frames of less than 64 bytes, unless it is already transferred because of lower value of Receive Threshold (e.g., RTC = 01).	0	R_W
5	Reserved	0	RO
4:3	RTC: Receive Threshold Control These two bits control the threshold level of the MTL Receive FIFO. Transfer (request) to DMA starts when the frame size within the MTL Receive FIFO is larger than the threshold. In addition, full frames with a length less than the threshold are transferred automatically. Note that value of 11 is not applicable if the configured Receive FIFO size is 128 bytes. These bits are valid only when the RSF bit is zero, and are ignored when the RSF bit is set to 1. 00 64 01 32 10 96 11 128	0	R_W
2	OSF: Operate on Second Frame When this bit is set, this bit instructs the DMA to process a second frame of Transmit data even before status for first frame is obtained.	0	R_W

Table 29-20 Register 6(Operation Mode Register)

Field	Description	Reset	Access
1	SR: Start/Stop Receive When this bit is set, the Receive process is placed in the Running state. The DMA attempts to acquire the descriptor from the Receive list and processes incoming frames. Descriptor acquisition is attempted from the current position in the list, which is the address set by DMA Register 3 or the position retained when the Receive process was previously stopped. If no descriptor is owned by the DMA, reception is suspended and Receive Buffer Unavailable (Register 5[7]) is set. The Start Receive command is effective only when reception has stopped. If the command was issued before setting DMA Register 3, DMA behavior is unpredictable. When this bit is cleared, RxDMA operation is stopped after the transfer of the current frame. The next descriptor position in the Receive list is saved and becomes the current position after the Receive process is restarted. The Stop Receive command is effective only when the Receive process is in either the Running (waiting for receive packet) or in the Suspended state.	0	R_W
0	Reserved	0	RO

29.5.2.1.8 Register 7(Interrupt Enable Register)

The Interrupt Enable register enables the interrupts reported by Register 5. Setting a bit to 1 enables a corresponding interrupt. After a hardware or software reset, all interrupts are disabled.

Table 29-21 Register 7(Interrupt enable Register)

Field	Description	Reset	Access
31	CRMBBFIE: Buffer B full status interrupt enable, set to enable the status to generate Abnormal interrupt.	0H	R_W
30	CRMBAFIE: Buffer A full status interrupt enable, set to enable the status to generate Abnormal interrupt.	0H	R_W
29:17	Reserved	0H	RO
16	NIE: Normal Interrupt Summary Enable When this bit is set, a normal interrupt is enabled. When this bit is reset, a normal interrupt is disabled. This bit enables the following bits: Register 5[0] Transmit Interrupt Register 5[2] Transmit Buffer Unavailable Register 5[6] Receive Interrupt Register 5[14] Early Receive Interrupt	0	R_W

Table 29-21 Register 7(Interrupt enable Register)

Field	Description	Reset	Access
15	AIE: Abnormal Interrupt Summary Enable When this bit is set, an Abnormal Interrupt is enabled. When this bit is reset, an Abnormal Interrupt is disabled. This bit enables the following bits Register 5[1] Transmit Process Stopped Register 5[3] Transmit Jabber Timeout Register 5[4] Receive Overflow Register 5[5] Transmit Underflow Register 5[7] Receive Buffer Unavailable Register 5[8] Receive Process Stopped Register 5[9] Receive Watchdog Timeout Register 5[10] Early Transmit Interrupt Register 5[13] Fatal Bus Error	0	R_W
14	ERE: Early Receive Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (Register 7[16]), Early Receive Interrupt is enabled. When this bit is reset, Early Receive Interrupt is disabled.	0	R_W
13	FBE: Fatal Bus Error Enable When this bit is set with Abnormal Interrupt Summary Enable (Register 7[15]), the Fatal Bus Error Interrupt is enabled. When this bit is reset, Fatal Bus Error Enable Interrupt is disabled.	0	RO
12:11	reserved	0	RO
10	ETE: Early Transmit Interrupt Enable When this bit is set with an Abnormal Interrupt Summary Enable (Register 7[15]), Early Transmit Interrupt is enabled. When this bit is reset, Early Transmit Interrupt is disabled.	0	R_W
9	RWE: Receive Watchdog Timeout Enable When this bit is set with Abnormal Interrupt Summary Enable (Register 7[15]), the Receive Watchdog Timeout Interrupt is enabled. When this bit is reset, Receive Watchdog Timeout Interrupt is disabled.	0	R_W
8	RSE: Receive Stopped Enable When this bit is set with Abnormal Interrupt Summary Enable (Register 7[15]), Receive Stopped Interrupt is enabled. When this bit is reset, Receive Stopped Interrupt is disabled.	0	R_W
7	RUE: Receive Buffer Unavailable Enable When this bit is set with Abnormal Interrupt Summary Enable (Register 7[15]), Receive Buffer Unavailable Interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable Interrupt is disabled.	0	R_W
6	RIE: Receive Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (Register 7[16]), Receive Interrupt is enabled. When this bit is reset, Receive Interrupt is disabled.	0	R_W
5	UNE: Underflow Interrupt Enable When this bit is set with Abnormal Interrupt Summary Enable (Register 7[15]), Transmit Underflow Interrupt is enabled. When this bit is reset, Underflow Interrupt is disabled.	0	R_W

Table 29-21 Register 7(Interrupt enable Register)

Field	Description	Reset	Access
4	OVE: Overflow Interrupt Enable When this bit is set with Abnormal Interrupt Summary Enable (Register 7[15]), Receive Overflow Interrupt is enabled. When this bit is reset, Overflow Interrupt is disabled	0	R_W
3	TJE: Transmit Jabber Timeout Enable When this bit is set with Abnormal Interrupt Summary Enable (Register 7[15]), Transmit Jabber Timeout Interrupt is enabled. When this bit is reset, Transmit Jabber Timeout Interrupt is disabled.	0	R_W
2	TUE: Transmit Buffer Unavailable Enable When this bit is set with Normal Interrupt Summary Enable (Register 7[16]), Transmit Buffer Unavailable Interrupt is enabled. When this bit is reset, Transmit Buffer Unavailable Interrupt is disabled.	0	R_W
1	TSE: Transmit Stopped Enable When this bit is set with Abnormal Interrupt Summary Enable (Register 7[15]), Transmission Stopped Interrupt is enabled. When this bit is reset, Transmission Stopped Interrupt is disabled.	0	R_W
0	TIE: Transmit Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (Register 7[16]), Transmit Interrupt is enabled. When this bit is reset, Transmit Interrupt is disabled.	0	R_W

The sbd_intr_o interrupt is generated as shown in **Figure 29-1**. It is asserted only when the TTI, MACPI, MACMI, or MACLI bits of the DMA Status register is asserted, or when the NIS/AIS Status bit is asserted and the corresponding Interrupt Enable bits (NIE/AIE) are enabled.

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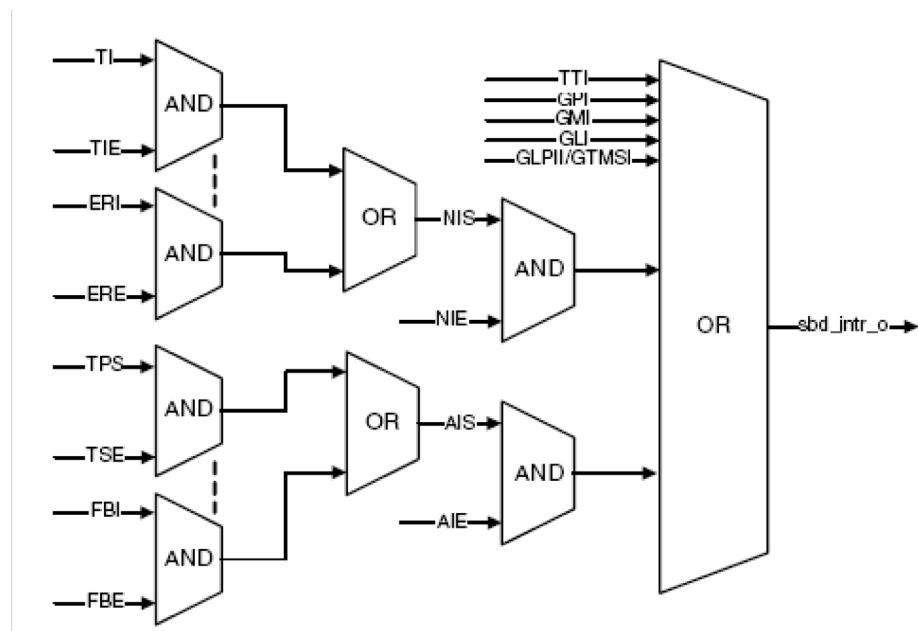


Figure 29-1 Sbd_Intr_o MACGenerate

29.5.2.1.9 Register 8(Missed Frame and Buffer Overflow Counter Register)

The DMA maintains two counters to track the number of missed frames during reception. This register reports the current value of the counter. The counter is used for diagnostic purposes. Bits[15:0] indicate missed frames because of the host buffer being unavailable. Bits[27:17] indicate missed frames because of buffer overflow conditions (MTL and MAC) and runt frames (good frames of less than 64 bytes) dropped by the MTL.

Table 29-22 Register 8(Missed Frame and Buffer Overflow Counter Register)

Field	Description	Reset	Access
31:29	Reserved	000H	RO
28	Overflow bit for FIFO Overflow Counter	0	R_SS_ RC
27:17	Indicates the number of frames missed by the application. This counter is incremented each time the MTL asserts the sideband signal mtl_rxoverflow_o. The counter is cleared when this register is read with mci_be_i[2] at 1'b1.	0	R_SS_ RC
16	Overflow bit for Missed Frame Counter	0	R_SS_ RC

Table 29-22 Register 8(Missed Frame and Buffer Overflow Counter Register)

Field	Description	Reset	Access
15:0	Indicates the number of frames missed by the controller because of the Host Receive Buffer being unavailable. This counter is incremented each time the DMA discards an incoming frame. The counter is cleared when this register is read with mci_be_i[0] at 1'b1.	0000H	R_SS_ RC

29.5.2.1.10 Register 9(Receive Interrupt Watchdog Timer Register)

This register, when written with non-zero value, enables the watchdog timer for Receive Interrupt (RI) (DMA Register 5[6]).

Table 29-23 Register 9(Receive Interrupt Watchdog Timer Register)

Field	Description	Reset	Access
31:8	Reserved	000000H	RO
7:0	RIWT: RI Watchdog Timer count Indicates the number of system clock cycles multiplied by 256 for which the watchdog timer is set. The watchdog timer gets triggered with the programmed value after the RxDMA completes the transfer of a frame for which the RI status bit is not set because of the setting in the corresponding descriptor RDES1[31]. When the watch-dog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when RI bit is set high because of automatic setting of RI as per RDES1[31] of any received frame.	00H	R_W

29.5.2.1.11 Register 18(Current Host Transmit Descriptor Register)

The Current Host Transmit Descriptor register points to the start address of the current Transmit Descriptor read by the DMA.

Table 29-24 Register 18(Current Host Transmit Descriptor Register)

Field	Description	Reset	Access
31:0	Host Receive Descriptor Address Pointer Cleared on Reset. Pointer updated by DMA during operation.	0000_0000 H	RO

29.5.2.1.12 Register 19(Current Host Receive Descriptor Register)

The Current Host Receive Descriptor register points to the start address of the current Receive Descriptor read by the DMA.

Table 29-25 Register 19(Current Host Receive Descriptor Register)

Field	Description	Reset	Access
31:0	Host Transmit Descriptor Address Pointer Cleared on Reset. Pointer updated by DMA during operation.	0000_0000 H	RO

29.5.2.1.13 Register 20(Current Host Transmit Buffer Address Register)

The Current Host Transmit Buffer Address register points to the current Transmit Buffer Address being read by the DMA.

Table 29-26 Register 20(Current Host Transmit Buffer Address Register)

Field	Description	Reset	Access
31:0	Host Transmit Descriptor Address Pointer Cleared on Reset. Pointer updated by DMA during operation.	0000_0000 H	RO

29.5.2.1.14 Register 21(Current Host Receive Buffer Address Register)

The Current Host Receive Buffer Address register points to the current Receive Buffer address being read by the DMA.

Table 29-27 Register 21(Current Host Receive Descriptor Register)

Field	Description	Reset	Access
31:0	Host Transmit Descriptor Address Pointer Cleared on Reset. Pointer updated by DMA during operation.	0000_0000 H	RO

29.5.2.1.15 Register 22(HW Feature Register)

This register indicates the presence of the optional features/functions of the core. It is useful for SW driver to dynamically enable or disable the programs related to the optional blocks.

Table 29-28 Register 22(HW Feature Register)

Field	Description	Access
31:25	Reserved	RO
24	Alternate (Enhanced Descriptor)	RO
23:22	Number of additional Tx Channels	RO
21:20	Number of additional Rx Channels	RO

Table 29-28 Register 22(HW Feature Register)

Field	Description	Access
19	RxFIFO > 2048 Bytes	RO
18	IP Checksum Offload (Type 2) in Rx	RO
17	IP Checksum Offload (Type 1) in Rx	RO
16	Checksum Offload in Tx	RO
15	AV Feature	RO
14	Energy Efficient Ethernet Feature	RO
13	IEEE 1588-2008 Advanced Time-Stamp	RO
12	IEEE 1588-2002 Time-Stamp	RO
11	RMON module	RO
10	PMT Magic Packet	RO
9	PMT Remote Wakeup	RO
8	SMA (MDIO) Interface	RO
7	Reserved	RO
6	PCS registers (TBI/SMACMII/RTBI PHY interface)	RO
5	Multiple MAC Address Registers	RO
4	HASH Filter	RO
3	Reserved	RO
2	Half-Duplex support	RO
1	1000 Mbps support	RO
0	10/100 Mbps support	RO

29.5.2.1.16 Register 23(continuous receive mode base address register)

In continuous rx data mode, this address is the base address of the 2-pingpong-buffer.

Table 29-29 Register 23(Continuous receive mode base address)

Field	Description	Reset	Access
31:2	CR MBA: The base address of 2-pingpong-buffer in continuous rx data mode.	000000H	R_W
1:0	Reserved	0H	RO

29.5.2.1.17 Register 24(continuous receive mode enable register)

Continuous rx data mode enable register.

Table 29-30 Register 24(Continuous receive mode base address)

Field	Description	Reset	Access
31	CRBA: Indicates which buffer is being written in continuous rx data mode, buffer A or buffer B.	1H	RO
30:1	Reserved	0H	RO
0	CRM: Continuous rx mode enable. This bit should be configured after all other continuous rx mode registers to enable the mode. Written '1' to enable the mode and reset registers for this mode. Written '0' to disable the mode and reset registers for this mode.	0H	R_W

29.5.2.1.18 Register 25(continuous receive mode current DMA write address register)

In continuous rx data mode, this address is the current write address of the DMA.

Table 29-31 Register 25(Continuous receive mode current DMA write address)

Field	Description	Reset	Access
31:0	CRMCDA: The current address of DMA write in continuous rx data mode.	0H	RO

29.5.2.1.19 Register 26(continuous receive mode buffer ceil address register)

In continuous rx data mode, this address is the max address of the rx buffer.

Table 29-32 Register 26(Continuous receive mode buffer ceil address)

Field	Description	Reset	Access
31:0	CRMBCA: The max address of rx data buffer in continuous rx data mode.	0H	R_W

29.5.2.1.20 Register 27(continuous receive mode buffer half size register)

In continuous rx data mode, this address is the half size of the rx buffer.

Table 29-33 Register 27(Continuous receive mode buffer half size register)

Field	Description	Reset	Access
31:16	Reserved	0H	RO
15:0	CRMBHS: The half size of rx data buffer in continuous rx data mode.	0H	R_W

29.5.2.1.21 Register 28(continuous receive mode buffer A desc start address register)

In continuous rx data mode, this address records the start address of the descriptors for buffer A after buffer A is full.

Table 29-34 Register 28(Continuous receive mode buffer A desc start address register)

Field	Description	Reset	Access
31:0	CRMBADSA: The start address of descriptors for buffer A in continuous rx data mode.	0H	RO

29.5.2.1.22 Register 29(continuous receive mode buffer B desc start address register)

In continuous rx data mode, this address records the start address of the descriptors for buffer B after buffer B is full.

Table 29-35 Register 29(Continuous receive mode buffer B desc start address register)

Field	Description	Reset	Access
31:0	CRMBBDSA: The start address of descriptors for buffer B in continuous rx data mode.	0H	RO

29.5.2.1.23 Register 30(continuous receive mode buffer A desc end address register)

In continuous rx data mode, this address records the end address of the descriptors for buffer A after buffer A is full.

Table 29-36 Register 30(Continuous receive mode buffer A desc end address register)

Field	Description	Reset	Access
31:0	CRMBADEA: The end address of descriptors for buffer A in continuous rx data mode.	0H	RO

29.5.2.1.24 Register 31(continuous receive mode buffer B desc end address register)

In continuous rx data mode, this address records the end address of the descriptors for buffer B after buffer B is full.

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Table 29-37 Register 31(Continuous receive mode buffer B desc end address register)

Field	Description	Reset	Access
31:0	CRMBBDEA: The end address of descriptors for buffer B in continuous rx data mode.	0H	RO

29.5.2.2 MAC Register Description

29.5.2.2.1 Register 0(MAC Configuration Register)

The MAC Configuration register establishes receive and transmit operating modes.

Table 29-38 Register 0(MAC Configuration Register)

Field	Description	Reset	Access
31	MAC_SPEED_10MBPS: Indicates that mac works at 10Mbps mode. To switch into this mode, clk_tx_i(also as clk_rx_i, which is a clock divided by 2 from clk_rmii_i in 100Mbps mode by default) should be configured as a clock divided by 20 from clk_rmii_i clock in system clock and power module(CPM).	0	RO
30:27	Reserved	0	RO
26	SFTERR: SMII Force Transmit Error When set, indicates to the PHY to force transmit error in the SMII frame being transmitted. This bit is reserved if SMII PHY port is not selected during core configuration.	0	RO
25	CST: CRC stripping for Type frames When set, the last 4 bytes (FCS) of all frames of Ether type (type field greater than 0x0600) are stripped and dropped before forwarding the frame to the application. This function is not valid when the MAC receiver has IP Checksum Engine (Type 1) enabled.	0	RO

Table 29-38 Register 0(MAC Configuration Register)

Field	Description	Reset	Access
24	TC: Transmit Configuration in RMACMII/SMACMII/SMII When set, this bit enables the transmission of duplex mode, link speed, and link up/down information to the PHY in the RMACMII/SMII/SMACMII ports. When this bit is reset, no such information is driven to the PHY. This bit is reserved (and RO) if neither RMACMII nor SMII nor SMACMII PHY port is selected during core configuration.	0	RO
23	WD: Watchdog Disable When this bit is set, the MAC disables the watchdog timer on the receiver, and can receive frames of up to 16,384 bytes. When this bit is reset, the MAC allows no more than 2,048 bytes (10,240 if JE is set high) of the frame being received and cuts off any bytes received after that.	0	RO
22	JD: Jabber Disable When this bit is set, the MAC disables the jabber timer on the transmitter, and can transfer frames of up to 16,384 bytes. When this bit is reset, the MAC cuts off the transmitter if the application sends out more than 2,048 bytes of data (10,240 if JE is set high) during transmission.	0	RO
21	BE: Frame Burst Enable When this bit is set, the MAC allows frame bursting during transmission in MACMII Half- Duplex mode. This bit is reserved (and RO) in 10/100 Mbps only or Full-Duplex-only configurations.	0	RO
20	JE: Jumbo Frame Enable When this bit is set, MAC allows Jumbo frames of 9,018 bytes (9,022 bytes for VLAN tagged frames) without reporting a giant frame error in the receive frame status.	0	RO
19:17	IFMAC: Inter-Frame MACap These bits control the minimum IFMAC between frames during transmission. 000 96 bit times 001 88 bit times 010 80 bit times ... 111 40 bit times In half-duplex mode, the minimum IFMAC can be configured for 64 bit times (IFMAC = 100) only. Lower values are not considered. In 1000-Mbps mode, the minimum IFMAC supported is 64 bit times (and above) in the MAC-CORE configuration and 80 bit times (and above) in other system configurations.	0	RO
16	DCRS: Disable Carrier Sense During Transmission When set high, this bit makes the MAC transmitter ignore the (MAC)MII CRS signal during frame transmission in Half-Duplex mode. This request results in no errors generated because of Loss of Carrier or No Carrier during such transmission. When this bit is low, the MAC transmitter generates such errors because of Carrier Sense and can even abort the transmissions. This bit is reserved (and RO) when the core is selected for Full-Duplex-only operation during core configuration.	0	RO

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Table 29-38 Register 0(MAC Configuration Register)

Field	Description	Reset	Access
15	PS: Port Select Selects between MACMII and MII: 0 MACMII (1000 Mbps) 1 MII (10/100 Mbps) This bit is read-only with the appropriate value in the 10/100 Mbps-only (always 1) or 1000 Mbps-only (always 0) configurations, and R_W in the default 10/100/1000 Mbps configuration.	0	RO
14	FES: Speed Indicates the speed in Fast Ethernet (MII) mode: 0 10 Mbps 1 100 Mbps This bit is reserved (RO) by default and is enabled only when RMII/SMII/RMACMII/SMACMII/RevMII is enabled during configuration. This bit generates link speed encoding when TC (Bit 24) is set in RMACMII/SMII/SMACMII mode. This signal is always driven for RevMII. If the core is configured with an RMII, SMII, SMACMII, or RMACMII PHY interface, this signal can optionally be driven as an output signal (mac_speed_o[0]). In addition, this bit is reserved when RMII is enabled during configuration without enabling the "Output Port for speed selection" option.	0	RO
13	DO: Disable Receive Own When this bit is set, the MAC disables the reception of frames when the gmii_txen_o is asserted in Half-Duplex mode. When this bit is reset, the MAC receives all packets that are given by the PHY while transmitting. This bit is not applicable if the MAC is operating in Full-Duplex mode. This bit is reserved (RO with default value) if the MAC is configured for Full-Duplex-only operation during configuration.	0	RO
12	LM: Loopback Mode When this bit is set, the MAC operates in loopback mode at MACMII/MII. The (MAC)MII Receive clock input (clk_rx_i) is required for the loopback to work properly, as the Transmit clock is not looped-back internally.	0	RO
11	DM: Duplex Mode When this bit is set, the MAC operates in a Full-Duplex mode where it can transmit and receive simultaneously. This bit is RO with default value of 1'b1 in Full-Duplex-only configuration.	0	RO

Table 29-38 Register 0(MAC Configuration Register)

Field	Description	Reset	Access
10	IPC: Checksum Offload When this bit is set, the MAC calculates the 16-bit one's complement of the one's complement sum of all received Ethernet frame payloads. It also checks whether the IPv4 Header checksum (assumed to be bytes 25–26 or 29–30 (VLAN-tagged) of the received Ethernet frame) is correct for the received frame and gives the status in the receive status word. The MAC core also appends the 16-bit checksum calculated for the IP header datagram payload (bytes after the IPv4 header) and appends it to the Ethernet frame transferred to the application (when Type 2 COE is deselected). When this bit is reset, this function is disabled. When Type 2 COE is selected, this bit, when set, enables IPv4 checksum checking for received frame payloads' TCP/UDP/ICMP headers. When this bit is reset, the COE function in the receiver is disabled and the corresponding PCE and IP HCE status bits are always cleared.	0	RO
9	DR: Disable Retry When this bit is set, the MAC attempts only 1 transmission. When a collision occurs on the MACMII/MII, the MAC ignores the current frame transmission and report a Frame Abort with excessive collision error in the transmit frame status. When this bit is reset, the MAC attempts retries based on the settings of BL (bit [6:5]). This bit is applicable only to Half-Duplex mode and is reserved (RO with default value) in Full-Duplex-only configuration.	0	RO
8	LUD: Link Up/Down Indicates whether the link is up or down during the transmission of configuration in RMACMII/SMACMII/SMII interface: 0 Link Down 1 Link Up This bit is reserved (RO with default value) and is enabled when RMACMII/SMACMII/SMII is enabled during configuration.	0	RO
7	ACS: Automatic Pad/CRC Stripping When this bit is set, the MAC strips the Pad/FCS field on incoming frames only if the length's field value is less than or equal to 1,500 bytes. All received frames with length field greater than or equal to 1,501 bytes are passed to the application without stripping the Pad/FCS field. When this bit is reset, the MAC passes all incoming frames to the Host unmodified.	0	RO

Table 29-38 Register 0(MAC Configuration Register)

Field	Description	Reset	Access
6:5	BL: Back-Off Limit The Back-Off limit determines the random integer number (r) of slot time delays (4,096 bit times for 1000 Mbps and 512 bit times for 10/100 Mbps) the MAC waits before rescheduling a transmission attempt during retries after a collision. This bit is applicable only to Half-Duplex mode and is reserved (RO) in Full-Duplex-only configuration. 00 k = min (n, 10) 01 k = min (n, 8) 10 k = min (n, 4) 11 k = min (n, 1) where n = retransmission attempt. The random integer r takes the value in the range $0 \leq r < 2^k$	0	RO
4	DC: Deferral Check When this bit is set, the deferral check function is enabled in the MAC. The MAC issues a Frame Abort status, along with the excessive deferral error bit set in the transmit frame status when the transmit state machine is deferred for more than 24,288 bit times in 10/100-Mbps mode. If the Core is configured for 1000 Mbps operation, or if the Jumbo frame mode is enabled in 10/100-Mbps mode, the threshold for deferral is 155,680 bits times. Deferral begins when the transmitter is ready to transmit, but is prevented because of an active CRS (carrier sense) signal on the MACMII/MII. Deferral time is not cumulative. If the transmitter defers for 10,000 bit times, then transmits, collides, backs off, and then has to defer again after completion of back-off, the deferral timer resets to 0 and restarts. When this bit is reset, the deferral check function is disabled and the MAC defers until the CRS signal goes inactive. This bit is applicable only in Half-Duplex mode and is reserved (RO) in Full-Duplex-only configuration.	0	RO
3	TE: Transmitter Enable When this bit is set, the transmit state machine of the MAC is enabled for transmission on the MACMII/MII. When this bit is reset, the MAC transmit state machine is disabled after the completion of the transmission of the current frame, and does not transmit any further frames.	0	RO
2	RE: Receiver Enable When this bit is set, the receiver state machine of the MAC is enabled for receiving frames from the MACMII/MII. When this bit is reset, the MAC receive state machine is disabled after the completion of the reception of the current frame, and does not receive any further frames from the MACMII/MII.	0	
1:0	1000 Mbps support	00	RO

29.5.2.2.2 Register 1(MAC Frame Filter)

The MAC Frame Filter register contains the filter controls for receiving frames. Some of the controls from this register go to the address check block of the MAC,

which performs the first level of address filtering. The second level of filtering is performed on the incoming frame, based on other controls such as Pass Bad Frames and Pass Control Frames.

Table 29-39 Register 1(MAC Frame Filter Register)

Field	Description	Reset	Access
31	RA: Receive All When this bit is set, the MAC Receiver module passes to the Application all frames received irrespective of whether they pass the address filter. The result of the SA/DA filtering is updated (pass or fail) in the corresponding bits in the Receive Status Word. When this bit is reset, the Receiver module passes to the Application only those frames that pass the SA/DA address filter.	0	R_W
30:11	Reserved	0000_000H	RO
10	HPF: Hash or Perfect Filter When set, this bit configures the address filter to pass a frame if it matches either the perfect filtering or the hash filtering as set by HMC or HUC bits. When low and if the HUC/HMC bit is set, the frame is passed only if it matches the Hash filter. This bit is reserved (and RO) if the Hash filter is not selected during core configuration.	0	R_W
9	SAF: Source Address Filter Enable The MAC core compares the SA field of the received frames with the values programmed in the enabled SA registers. If the comparison matches, then the SAMatch bit of RxStatus Word is set high. When this bit is set high and the SA filter fails, the MAC drops the frame. When this bit is reset, then the MAC Core forwards the received frame to the application and with the updated SA Match bit of the RxStatus depending on the SA address comparison.	0	R_W
8	SAIF: SA Inverse Filtering When this bit is set, the Address Check block operates in inverse filtering mode for the SA address comparison. The frames whose SA matches the SA registers are marked as failing the SA Address filter. When this bit is reset, frames whose SA does not match the SA registers are marked as failing the SA Address filter.	0	R_W
7:6	PCF: Pass Control Frames These bits control the forwarding of all control frames (including unicast and multicast PAUSE frames). Note that the processing of PAUSE control frames depends only on RFE of Flow Control Register[2]. 00 MAC filters all control frames from reaching the application. 01 MAC forwards all control frames except PAUSE control frames to application even if they fail the Address filter. 10 MAC forwards all control frames to application even if they fail the Address Filter. 11 MAC forwards control frames that pass the Address Filter.	0	R_W

Table 29-39 Register 1(MAC Frame Filter Register)

Field	Description	Reset	Access
5	DBF: Disable Broadcast Frames When this bit is set, the AFM module filters all incoming broadcast frames. In addition, it overrides all other filter settings. When this bit is reset, the AFM module passes all received broadcast frames.	0	R_W
4	PM: Pass All Multicast When set, this bit indicates that all received frames with a multicast destination address (first bit in the destination address field is '1') are passed. When reset, filtering of multicast frame depends on HMC bit.	0	R_W
3	DAIF: DA Inverse Filtering When this bit is set, the Address Check block operates in inverse filtering mode for the DA address comparison for both unicast and multicast frames. When reset, normal filtering of frames is performed.	0	R_W
2	HMC: Hash Multicast When set, MAC performs destination address filtering of received multicast frames according to the hash table. When reset, the MAC performs a perfect destination address filtering for multicast frames, that is, it compares the DA field with the values programmed in DA registers.	0	R_W
1	HUC: Hash Unicast When set, MAC performs destination address filtering of unicast frames according to the hash table. When reset, the MAC performs a perfect destination address filtering for unicast frames, that is, it compares the DA field with the values programmed in DA registers.	0	R_W
0	PR: Promiscuous Mode When this bit is set, the Address Filter module passes all incoming frames regardless of its destination or source address. The SA/DA Filter Fails status bits of the Receive Status Word are always cleared when PR is set.	0	R_W

29.5.2.2.3 Register 2(Hash Table High Register)

The 64-bit Hash table is used for group address filtering. For hash filtering, the contents of the destination address in the incoming frame is passed through the CRC logic, and the upper 6 bits of the CRC register are used to index the contents of the Hash table. The most significant bit determines the register to be used (Hash Table High/Hash Table Low), and the other 5 bits determine which bit within the register. A hash value of 5b'00000 selects Bit 0 of the selected register, and a value of 5b'11111 selects Bit 31 of the selected register.

For example, if the DA of the incoming frame is received as 0x1F52419CB6AF (0x1F is the first byte received on MACMII interface), then the internally calculated 6-bit Hash value is 0x2C and the HTH register bit[12] is checked for filtering. If the

DA of the incoming frame is received as 0xA00A98000045, then the calculated 6-bit Hash value is 0x07 and the HTL register bit[7] is checked for filtering.

If the corresponding bit value of the register is 1'b1, the frame is accepted. Otherwise, it is rejected. If the PM (Pass All Multicast) bit is set in Register 1, then all multicast frames are accepted regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (MAC)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table High/Low registers are written to. Please note that consecutive writes to these register should be performed only after at least 4 clock cycles in the destination clock domain when double-synchronization is enabled.

The Hash Table High register contains the higher 32 bits of the Hash table.

Table 29-40 Register 2(Hash Table High Register)

Field	Description	Reset	Access
31:0	HTH: Hash Table High This field contains the upper 32 bits of Hash table.	0000_0000 H	R_W

29.5.2.2.4 Register 3(Hash Table Low Register)

The Hash Table Low register contains the lower 32 bits of the Hash table.

Table 29-41 Register 3(Hash Table Low Register)

Field	Description	Reset	Access
31:0	HTL: Hash Table Low This field contains the lower 32 bits of Hash table.	0000_0000 H	R_W

29.5.2.2.5 Register 4(MACMII Address Register)

The MACMII Address register controls the management cycles to the external PHY through the management interface.

Table 29-42 Register 4(MACMII Address Register)

Field	Description	Reset	Access
31:16	Reserved	0000H	RO

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Table 29-42 Register 4(MACMII Address Register)

Field	Description	Reset	Access
15:11	PA: Physical Layer Address This field tells which of the 32 possible PHY devices are being accessed. For RevMII, this field gives the PHY Address of the RevMII module.		
10:6	MACR: MACMII Register These bits select the desired MACMII register in the selected PHY device.		R_W
5:2	CR: CSR Clock Range The CSR Clock Range selection determines the frequency of the MDC clock as per the clk_csr_i frequency used in your design. The suggested range of clk_csr_i frequency applicable for each value below (when Bit[5] = 0) ensures that the MDC clock is approximately between the frequency range 1.0 MHz - 2.5 MHz. Selection clk_csr_i MDC Clock 0000 60-100 MHz clk_csr_i/42 0001 100-150 MHz clk_csr_i/62 0010 20-35 MHz clk_csr_i/16 0011 35-60 MHz clk_csr_i/26 0100 150-250 MHz clk_csr_i/102 0101 250-300 MHz clk_csr_i/124 0110, 0111 Reserved When bit 5 is set, you can achieve MDC clock of frequency higher than the IEEE 802.3 specified frequency limit of 2.5 MHz and program a clock divider of lower value. For example, when clk_csr_i is of frequency 100 Mhz and you program these bits as "1010", then the resultant MDC clock is of 12.5 Mhz which is outside the limit of IEEE 802.3 specified range. Please program the values given below only if the interfacing chips supports faster MDC clocks. Selection MDC Clock 1000 clk_csr_i/4 1001 clk_csr_i/6 1010 clk_csr_i/8 1011 clk_csr_i/10 1100 clk_csr_i/12 1101 clk_csr_i/14 1110 clk_csr_i/16 1111 clk_csr_i/18 These bits are not used for accessing RevMII.		R_W
1	MACW: MACMII Write When set, this bit tells the PHY/RevMII that this is a Write operation using the MACMII Data register. If this bit is not set, this is a Read operation, placing the data in the MACMII Data register.		R_W

Table 29-42 Register 4(MACMII Address Register)

Field	Description	Reset	Access
0	MACB: MACMII Busy This bit should read a logic 0 before writing to Register 4 and Register 5. During a PHY/RevMII register access, the Software sets this bit to 1'b1 to indicate that a Read or Write access is in progress. The Register 5 is invalid until this bit is cleared by the MAC. Therefore, Register 5 (MACMII Data) should be kept valid until the MAC clears this bit during a PHY Write operation. Similarly for read operation, Register 5 contents are not valid until this bit is cleared. The subsequent read/write operation should happen only after the previous operation is complete.	0000_0000 H	R_W

29.5.2.2.6 Register 5(MACMII Data Register)

The MACMII Data register stores Write data to be written to the PHY register located at the address specified in Register 4. Register 5 also stores Read data from the PHY register located at the address specified by Register 4.

Table 29-43 Register 5(MACMII Data Register)

Field	Description	Reset	Access
31:16	Reserved	0000H	RO
15:0	MACD: MACMII Data This contains the 16-bit data value read from the PHY/RevMII after a Management Read operation or the 16-bit data value to be written to the PHY/RevMII before a Management Write operation.	0000H	R_W

29.5.2.2.7 Register 6(Flow Control Register)

The Flow Control register controls the generation and reception of the Control (Pause Command) frames by the MAC's Flow control module. A Write to a register with the Busy bit set to '1' triggers the Flow Control block to generate a Pause Control frame. The fields of the control frame are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control frame. The Busy bit remains set until the control frame is transferred onto the cable. The Host must make sure that the Busy bit is cleared before writing to the register.

Table 29-44 Register 6(Flow Control Register)

Field	Description	Reset	Access
31:16	PT: Pause Time This field holds the value to be used in the Pause Time field in the transmit control frame. If the Pause Time bits is configured to be double-synchronized to the (MAC)MII clock domain, then consecutive writes to this register should be performed only after at least 4 clock cycles in the destination clock domain.	0000H	R_W
15:8	Reserved	0000H	RO
7	DZPQ: Disable Zero-Quanta Pause When set, this bit disables the automatic generation of Zero-Quanta Pause Control frames on the de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i/mti_flowctrl_i). When this bit is reset, normal operation with automatic Zero-Quanta Pause Control frame generation is enabled.	0	R_W
6	Reserved	0	RO
5:4	PLT: Pause Low Threshold This field configures the threshold of the PAUSE timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of PAUSE Frame. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot-times), and PLT = 01, then a second PAUSE frame is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256 – 28) slot-times after the first PAUSE frame is transmitted. Selection Threshold 00 Pause time minus 4 slot times 01 Pause time minus 28 slot times 10 Pause time minus 144 slot times 11 Pause time minus 256 slot times The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the MACMII/MII interface.	0	R_W
3	UP: Unicast Pause Frame Detect When this bit is set, the MAC detects the Pause frames with the station's unicast address specified in MAC Address0 High Register and MAC Address0 Low Register, in addition to the detecting Pause frames with the unique multicast address. When this bit is reset, the MAC detects only a Pause frame with the unique multicast address specified in the 802.3x standard.	0	R_W
2	RFE: Receive Flow Control Enable When this bit is set, the MAC decodes the received Pause frame and disable its transmitter for a specified (Pause Time) time. When this bit is reset, the decode function of the Pause frame is disabled.	0	R_W

Table 29-44 Register 6(Flow Control Register)

Field	Description	Reset	Access
1	TFE: Transmit Flow Control Enable In Full-Duplex mode, when this bit is set, the MAC enables the flow control operation to transmit Pause frames. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause frames. In Half-Duplex mode, when this bit is set, the MAC enables the back-pressure operation. When this bit is reset, the back-pressure feature is disabled.	0	R_W
0	FCB/BPA: Flow Control Busy/Backpressure Activate This bit initiates a Pause Control frame in Full-Duplex mode and activates the backpressure function in Half-Duplex mode if TFE bit is set. In Full-Duplex mode, this bit should be read as 1'b0 before writing to the Flow Control register. To initiate a Pause control frame, the Application must set this bit to 1'b1. During a transfer of the Control Frame, this bit continues to be set to signify that a frame transmission is in progress. After the completion of Pause control frame transmission, the MAC resets this bit to 1'b0. The Flow Control register should not be written to until this bit is cleared. In Half-Duplex mode, when this bit is set (and TFE is set), then backpressure is asserted by the MAC Core. During backpressure, when the MAC receives a new frame, the transmitter starts sending a JAM pattern resulting in a collision. This control register bit is logically OR'ed with the mti_flowctrl_i input signal for the backpressure function. When the MAC is configured to Full-Duplex mode, the BPA is automatically disabled.	0	R_WS_SC(FCB)

29.5.2.2.8 Register 7(VLAN Tag Register)

The VLAN Tag register contains the IEEE 802.1Q VLAN Tag to identify the VLAN frames. The MAC compares the 13th and 14th bytes of the receiving frame (Length/Type) with 16?h8100, and the following 2 bytes are compared with the VLAN tag; if a match occurs, it sets the received VLAN bit in the receive frame status. The legal length of the frame is increased from 1518 bytes to 1522 bytes.

If the VLAN Tag register is configured to be double-synchronized to the (MAC)MII clock domain, then consecutive writes to these register should be performed only after at least 4 clock cycles in the destination clock domain.

Table 29-45 Register 7(VLAN Tag Register)

Field	Description	Reset	Access
31:17	Reserved	0000H	RO

Table 29-45 Register 7(VLAN Tag Register)

Field	Description	Reset	Access
16	ETV: Enable 12-Bit VLAN Tag Comparison When this bit is set, a 12-bit VLAN identifier, rather than the complete 16-bit VLAN tag, is used for comparison and filtering. Bits[11:0] of the VLAN tag are compared with the corresponding field in the received VLAN-tagged frame. When this bit is reset, all 16 bits of the received VLAN frame's fifteenth and sixteenth bytes are used for comparison.	0	R/W
15:0	VL: VLAN Tag Identifier for Receive Frames This contains the 802.1Q VLAN tag to identify VLAN frames, and is compared to the fifteenth and sixteenth bytes of the frames being received for VLAN frames. Bits[15:13] are the User Priority, Bit[12] is the Canonical Format Indicator (CFI) and bits[11:0] are the VLAN tag's VLAN Identifier (VID) field. When the ETV bit is set, only the VID (Bits[11:0]) is used for comparison. If VL (VL[11:0] if ETV is set) is all zeros, the MAC does not check the fifteenth and sixteenth bytes for VLAN tag comparison, and declares all frames with a Type field value of 0x8100 to be VLAN frames.	0000H	R_W

29.5.2.2.9 Register 8(Debug Register)

This debug register gives the status of all the main modules of the transmit and receive data-paths and the FIFOs. An all-zero status indicates that the MAC core is in idle state (and FIFOs are empty) and no activity is going on in the data-paths.

Table 29-46 Register8(Debug Register)

Field	Description	Reset	Access
31:26	Reserved	0000H	RO
25	When high, it indicates that the MTL TxStatus FIFO is full and hence the MTL cannot accept any more frames for transmission. This bit is reserved in the MAC-AHB and MAC-DMA configurations.	0	RO
24	When high, it indicates that the MTL TxFIFO is not empty and has some data left for transmission.	0	RO
23	Reserved	0	RO
22	When high, it indicates that the MTL TxFIFO Write Controller is active and transferring data to the TxFIFO.	0	RO
21:20	This indicates the state of the TxFIFO read Controller: 00 IDLE state 01 READ state (transferring data to MAC transmitter) 10 Waiting for TxStatus from MAC transmitter 11 Writing the received TxStatus or flushing the TxFIFO	0	RO
19	When high, it indicates that the MAC transmitter is in PAUSE condition (in full-duplex only) and hence does not schedule any frame for transmission	0	RO

Table 29-46 Register8(Debug Register)

Field	Description	Reset	Access
18:17	This indicates the state of the MAC Transmit Frame Controller module: 00 IDLE state 01 Waiting for Status of previous frame or IFMAC/backoff period to be over 10 MACGenerating and transmitting a PAUSE control frame (in full duplex mode) 11 Transferring input frame for transmission	0	RO
16	When high, it indicates that the MAC MACMII/MII transmit protocol engine is actively transmitting data and not in IDLE state.	0	RO
15:10	Reserved	0	RO
9:8	This gives the status of the RxFIFO Fill-level: 00 RxFIFO Empty 01 RxFIFO fill-level below flow-control de-activate threshold 10 RxFIFO fill-level above flow-control activate threshold 11 RxFIFO Full	0	RO
7	Reserved	0	RO
6:5	It gives the state of the RxFIFO read Controller: 00 IDLE state 01 Reading frame data 10 Reading frame status (or time-stamp) 11 Flushing the frame data and Status	0	RO
4	When high, it indicates that the MTL RxFIFO Write Controller is active and transferring a received frame to the FIFO.	0	RO
3	Reserved	0	RO
2:1	When high, it indicates the active state of the small FIFO Read and Write controllers respectively of the MAC receive Frame Controller module	0	RO
0	When high, it indicates that the MAC MACMII/MII receive protocol engine is actively receiving data and not in IDLE state.	0	RO

29.5.2.2.10 Register 9(Remote Wake-up Frame Filter Register)

The register `wkupfmfilter_reg`, at address (028H), loads the Wake-up Frame Filter register. To load values in a Wake-up Frame Filter register, the entire register (`wkupfmfilter_reg`) must be written. The `wkupfmfilter_reg` register is loaded by sequentially loading the eight register values in address (028) for `wkupfmfilter_reg0`, `wkupfmfilter_reg1`, ... `wkupfmfilter_reg7`, respectively. The `wkupfmfilter_reg` register is read in a similar way.

wkupfmfilter_reg0	Filter 0 Byte Mask							
wkupfmfilter_reg1	Filter 1 Byte Mask							
wkupfmfilter_reg2	Filter 2 Byte Mask							
wkupfmfilter_reg3	Filter 3 Byte Mask							
wkupfmfilter_reg4	RSVD	Filter 3 Command	RSVD	Filter 2 Command	RSVD	Filter 1 Command	RSVD	Filter 0 Command
wkupfmfilter_reg5	Filter 3 Offset		Filter 2 Offset		Filter 1 Offset		Filter 0 Offset	
wkupfmfilter_reg6	Filter 1 CRC - 16				Filter 0 CRC - 16			
wkupfmfilter_reg7	Filter 3 CRC - 16				Filter 2 CRC - 16			

Figure 29-1 Wake-up Frame Filter Register

1. Filter i Byte Mask

The Filter i Byte Mask register defines the bytes of the frame that are examined by filter i (0, 1, 2, and 3) in order to determine whether or not a frame is a wake-up frame. The MSB (thirty-first bit) must be zero. The bit j [30:0] is the Byte Mask. If bit j (byte number) of the Byte Mask is set, then the CRC block processes the Filter i Offset + j of the incoming frame; otherwise Filter i Offset + j is ignored.

2. Filter i Command

The 4-bit Filter i Command controls the filter i operation. The bit 3 specifies the address type, defining the destination address type of the pattern. When the bit is set, the pattern applies to only multicast frames; when the bit is reset, the pattern applies only to unicast frame. Bit 2 and Bit 1 are reserved. Bit 0 is the enable for filter i. If bit 0 is not set, filter i is disabled.

3. Filter i Offset

This Filter i Offset register defines the offset (within the frame) from which the filter i examines the frames. This 8-bit pattern-offset is the offset for the filter i first byte to be examined. The minimum allowed offset is 12, which refers to the 13th byte of the frame. The offset value 0 refers to the first byte of the frame.

4. Filter i CRC-16

This Filter i CRC-16 register contains the CRC_16 value calculated from

the pattern, as well as the byte mask programmed to the wake-up filter register block.

29.5.2.2.11 Register 9(PMT Control and Status Register)

The PMT CSR programs the request wake-up events and monitors the wake-up events.

Table 29-47 Register9(PMT Control and Status Register)

Field	Description	Reset	Access
31	Wake-Up Frame Filter Register Pointer Reset When set, resets the remote wake-up frame filter register pointer to 3'b000. It is automatically cleared after 1 clock cycle.	0000H	R_WS_SC
30:10	Reserved	0000_00H	RO
9	MAClobal Unicast When set, enables any unicast packet filtered by the MAC (DAF) address recognition to be a wake-up frame.	0	R_W
8:7	Reserved	0	RO
6	Wake-Up Frame Received When set, this bit indicates the power management event is generated because of the reception of a wake-up frame. This bit is cleared by a Read into this register.	0	R_SS_RC
5	Magic Packet Received When set, this bit indicates that the power management event is generated because of the reception of a magic packet. This bit is cleared by a Read into this register.	0	R_SS_RC
4:3	Reserved	00	RO
2	Wake-Up Frame Enable When set, enables generation of a power management event because of wake-up frame reception.	0	R_W
1	Magic Packet Enable When set, enables generation of a power management event because of magic packet reception.	0	R_W
0	Power Down When set, all received frames are dropped. This bit is cleared automatically when a magic packet or wake-up frame is received, and power-down mode is disabled. Frames, that are received after this bit is cleared, are forwarded to the application. This bit must only be set when the Magic Packet Enable, MAClobal Unicast, or Wake- Up Frame Enable bit is set high.	0	R_WS_SC

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29.5.2.2.12 Register 11(Interrupt Status Register)

The Interrupt Status register contents identify the events in the MAC-CORE that can generate interrupt.

Table 29-48 Register11(Interrupt Status Register)

Field	Description	Reset	Access
31:1 1	Reserved	0000H	RO
10	Reserved.	0	RO
9	Reserved.	0	RO/ R_SS_ RC
8	Reserved	0	RO
7	MMC Receive Checksum Offload Interrupt Status This bit is set high whenever an interrupt is generated in the MMC Receive Checksum Offload Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared. This bit is valid only when you select the optional MMC module and Checksum Offload Engine (Type 2) during configuration.	0	RO
6	MMC Transmit Interrupt Status This bit is set high whenever an interrupt is generated in the MMC Transmit Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared. This bit is valid only when you select the optional MMC module during configuration.	0	RO
5	MMC Receive Interrupt Status This bit is set high whenever an interrupt is generated in the MMC Receive Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared. This bit is valid only when you select the optional MMC module during configuration.	0	RO
4	MMC Interrupt Status This bit is set high whenever any of bits 7:5 is set high and cleared only when all of these bits are low. This bit is valid only when you select the optional MMC module during configuration.	0	RO
3	PMT Interrupt Status This bit is set whenever a Magic packet or Wake-on-LAN frame is received in Power- Down mode See bits 5 and 6 in the “PMT Control and Status Register 29.5.2.2.11 ”. This bit is cleared when both bits[6:5] are cleared because of a read operation to the PMT Control and Status register. This bit is valid only when you select the optional PMT module during configuration.	0	RO

Table 29-48 Register11(Interrupt Status Register)

Field	Description	Reset	Access
2	PMT Interrupt Status This bit is set whenever a Magic packet or Wake-on-LAN frame is received in Power- Down mode See bits 5 and 6 in the “PMT Control and Status Register 29.5.2.2.11 ” . This bit is cleared when both bits[6:5] are cleared because of a read operation to the PMT Control and Status register. This bit is valid only when you select the optional PMT module during configuration.	0	RO
1	PCS Link Status Changed This bit is set because of any change in Link Status in the TBI/RTBI/SMACMII PHY interface. This bit is cleared when the user makes a read operation the AN Status register. This bit is valid only when you select the optional TBI/RTBI/SMACMII PHY interface during configuration and operation.	0	RO
0	RMACMII/SMII Interrupt Status This bit is set because of any change in value of the Link Status of RMACMII/SMII interface. This bit is cleared when the user makes a read operation the SMACMII/RMACMII/SMII Status register. This bit is valid only when you select the optional RMACMII/SMII PHY interface during configuration and operation.	0	RO

29.5.2.2.13 Register 12(Interrupt Mask Register)

The Interrupt Mask Register bits enables the user to mask the interrupt signal because of the corresponding event in the Interrupt Status Register. The interrupt signal is `sbd_intr_o` in MAC-AHB, MAC-AXI, and MAC-DMA configuration while the interrupt signal is `mci_intr_o` in the MAC-MTL and MAC-CORE configuration.

Table 29-49 Register12(Interrupt Mask Register)

Field	Description	Reset	Access
31:1 1	Reserved	0000H	RO
10	Reserved.	0	RO
9	Reserved.	0	RO/ R_SS_ RC
8:4	Reserved	0	RO
7	PMT Interrupt Mask When set, this bit disables the assertion of the interrupt signal because of the setting of PMT Interrupt Status bit in Register 14 29.5.2.2.12 (Interrupt Status Register).	0	RO

Table 29-49 Register12(Interrupt Mask Register)

Field	Description	Reset	Access
6	PCS AN Completion Interrupt Mask When set, this bit disables the assertion of the interrupt signal because of the setting of PCS Auto-negotiation complete bit in Register 14 29.5.2.2.12 (Interrupt Status Register) caused because of the completion of Auto-negotiation event.	0	RO
5	PCS Link Status Interrupt Mask When set, this bit disables the assertion of the interrupt signal because of the setting of the PCS Link-status changed bit in Register 14 29.5.2.2.12 (Interrupt Status Register) caused because of change in link-status event.	0	RO
4	RMACMII/SMII Interrupt Mask When set, this bit disables the assertion of the interrupt signal because of the setting of RMACMII/SMII Interrupt Status bit in Register 14 29.5.2.2.12 (Interrupt Status Register).	0	RO

29.5.2.2.14 Register 13(MAC Address 0 High Register)

The MAC Address0 High register holds the upper 16 bits of the 6-byte first MAC address of the station. Note that the first DA byte that is received on the (MAC)MII interface corresponds to the LS Byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 is the first byte) on the (MAC)MII as the destination address, then the MacAddress0 Register [47:0] is compared with 0x665544332211.

If the MAC address registers are configured to be double-synchronized to the (MAC)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address Low Register (Register 17) are written to. Please note that consecutive writes to this Address Low Register should be performed only after at least 4 clock cycles in the destination clock domain for proper synchronization updates.

Table 29-50 Register13(MAC Address0 High Register)

Field	Description	Reset	Access
31	MO: Always 1.	1	RO
30:16	Reserved	0000H	RO
15:0	A[47:32]: MAC Address0 [47:32] This field contains the upper 16 bits (47:32) of the 6-byte first MAC address. This is used by the MAC for filtering for received frames and for inserting the MAC address in the Transmit Flow Control (PAUSE) Frames.	FFFFH	R_W

29.5.2.2.15 Register 14(MAC Address 0 Low Register)

The MAC Address0 Low register holds the lower 32 bits of the 6-byte first MAC address of the station.

Table 29-51 Register14(MAC Address0 Low Register)

Field	Description	Reset	Access
31:0	A[31:0]: MAC Address0 [31:0] A[31:0]: MAC Address0 [31:0] This field contains the lower 32 bits of the 6-byte first MAC address. This is used by the MAC for filtering for received frames and for inserting the MAC address in the Transmit Flow Control (PAUSE) Frames.	FFFF_FFF_FH	R_W

29.5.2.2.16 Register 15(MAC Address 1 High Register)

The MAC Address1 High register holds the upper 16 bits of the 6-byte second MAC address of the station.

If the MAC address registers are configured to be double-synchronized to the (MAC)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address Low Register (Register 19) are written to. Consecutive writes to this Address Low Register must be performed only after at least 4 clock cycles in the destination clock domain for proper synchronization updates.

Table 29-52 Register15(MAC Address1 High Register)

Field	Description	Reset	Access
31	AE: Address Enable When this bit is set, the Address filter module uses the second MAC address for perfect filtering. When reset, the address filter module ignores the address for filtering.	0	R_W
30	SA: Source Address When this bit is set, the MAC Address1[47:0] is used to compare with the SA fields of the received frame. When this bit is reset, the MAC Address1[47:0] is used to compare with the DA fields of the received frame.	0	R_W

Table 29-52 Register15(MAC Address1 High Register)

Field	Description	Reset	Access
29:2 4	MBC: Mask Byte Control These bits are mask control bits for comparison of each of the MAC Address bytes. When set high, the MAC core does not compare the corresponding byte of received DA/SA with the contents of MAC Address1 registers. Each bit controls the masking of the bytes as follows: Bit 29 Register 18[15:8] Bit 28 Register 18[7:0] Bit 27 Register 19[31:24] ... Bit 24 Register 19[7:0]	000000	R_W
23:1 6	Reserved	00H	RO
15:0	A[47:32]: MAC Address1 [47:32] This field contains the upper 16 bits (47:32) of the 6-byte second MAC address.	FFFFH	R_W

29.5.2.2.17 Register 16(MAC Address 1 Low Register)

The MAC Address1 Low register holds the lower 32 bits of the 6-byte second MAC address of the station.

Table 29-53 Register16(MAC Address1 Low Register)

Field	Description	Reset	Access
31:0	A[31:0]: MAC Address1 [31:0] This field contains the lower 32 bits of the 6-byte second MAC address. The content of this field is undefined until loaded by the Application after the initialization process.	FFFF_FFFFH	R_W

Note

- The descriptions for registers 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, and 46 (MAC Address2 High Register through MAC Address15 High Register) are the same as for the Register 18 MAC Address1 High Register.

- The descriptions for registers 21, 23, 25, 27, 29, 31, 33, 35, 37, 38, 41, 43, 45, and 47 (MAC Address2 Low Register through MAC Address15 Low Register) are the same as for the Register 19 MAC Address1 Low Register.
- The descriptions for registers 512, 514, 516, 518, 520, 522, 524, 526, 528, 530, 532, 534, 536, 538, 540, and 542 (MAC Address16 High Register through MAC Address31 High Register) are the same as for the Register 18 MAC Address1 High Register.
- The descriptions for registers 513, 515, 517, 519, 521, 523, 525, 527, 529, 531, 533, 535, 537, 539, 541, and 543 (MAC Address16 Low Register through MAC Address31 Low Register) is the same as for the Register 19 MAC Address1 Low Register.

29.5.2.2.18 Register 54(SMACMII/RMACMII/SMII Status Register)

The SMACMII/RMACMII/SMII Status register indicates the status signals received by the SMACMII/RMACMII/SMII (whichever is selected at reset) from the PHY. This register is optional and is present only when the core is configured for RMACMII/SMACMII PHY interfaces.

Table 29-54 Register54(SMACMII/RMACMII/SMII Register)

Field	Description	Reset	Access
31:6	Reserved	000_0000H	RO
5	False Carrier Detected Indicates whether SMII PHY detected false carrier (1'b1). Reserved when the core is configured for SMACMII/RMACMII PHY interfaces.	0	RO
4	Jabber Timeout Indicates whether there was jabber timeout error (1'b1) in received frame. Reserved when the core is configured for SMACMII/RMACMII PHY interfaces.	0	RO
3	Link Status Indicates whether the link is up (1'b1) or down (1'b0).	0	RO

Table 29-54 Register54(SMACMII/RMACMII/SMII Register)

Field	Description	Reset	Access
2:1	Link Speed Indicates the current speed of the link: 00 2.5 MHz 01 25 MHz 10 125 MHz Bit 2 is reserved when the core is configured for SMII PHY interface.	0	RO
0	Link Mode Indicates the current mode of operation of the link: 1'b0 Half-Duplex mode 1'b1 Full-Duplex mode	0	RO

29.6 Programming MACuide

29.6.1 Initializing DMA

This initialization sequence is used for MAC configuration.

Perform the following steps to initialize the DMA:

1. Provide a software reset. This resets all of the MAC internal registers and logic. (DMA Register 0 (Bus Mode Register) - bit 0).
2. Wait for the completion of the reset process (poll bit 0 of the DMA Register 0 (Bus Mode Register), which is only cleared after the reset operation is completed).
3. Program the following fields to initialize the Bus Mode Register by setting values in DMA Register 0 (Bus Mode Register):
 - c. Mixed Burst and AAL (only if MAC-AHB/MAC-AXI configuration is selected)
 - d. Fixed burst or undefined burst
 - e. Burst length values and burst mode values
 - f. Descriptor Length (only valid if Ring Mode is used)
 - g. Tx and Rx DMA Arbitration scheme
4. Create a proper descriptor chain for transmit and receive. In addition, ensure that the receive descriptors are owned by DMA (bit 31 of descriptor should be set). When OSF mode is used, at least two descriptors are required.
5. Make sure that your software creates three or more different transmit or receive descriptors in the chain before reusing any of the descriptors.
6. Initialize receive and transmit descriptor list address with the base address of the transmit and receive descriptor (Register 3 (Receive Descriptor List Address Register) and Register 4 (Transmit Descriptor List Address Register) respectively).
7. Program the following fields to initialize the mode of operation in Register 6

(Operation Mode Register)

- a. Tx and Rx DMA Arbitration scheme
 - b. Receive and Transmit Threshold Control (RTC and TTC)
 - c. Hardware Flow Control enable
 - d. Flow Control Activation and De-activation thresholds for MTL Receive and Transmit FIFO (RFA and RFD)
 - e. Error Frame and undersized good frame forwarding enable
 - f. OSF Mode
8. Mixed Burst and AAL (only if MAC-AHB/MAC-AXI configuration is selected)
 9. Clear the interrupt requests, by writing to those bits of the status register (interrupt bits only) that are set. For example, by writing 1 into bit 16, the normal interrupt summary clears this bit (DMA Register 5 (Status Register)).
 10. Enable the interrupts by programming the Register 7 (Interrupt Enable Register).
 11. Start the Receive and Transmit DMA by setting SR (bit 1) and ST (bit 13) of the control register (DMA Register 6 (Operation Mode Register)).

29.6.2 Initializing MAC

The following MAC Initialization operations can be performed after DMA initialization. If the MAC initialization is done before the DMA is set-up, then enable the MAC receiver (last step below) only after the DMA is active. Otherwise, received frames fill the Rx FIFO and overflow. Steps (1) and (2) are to be followed if the TBI, SMACMII, or RTBI PHY interface is enabled, otherwise follow steps (3) and (4).

1. Program the MAC Register 48 (AN Control Register) to enable Auto-negotiation ANE (bit-12). Setting ELE (bit-14) of this register enables the PHY to loop back the transmit data and RAN (bit-9) can be set to restart Auto negotiation (only for TBI, SMACMII, and RTBI PHY interfaces).
2. Check the MAC Register 49 (AN Status Register) for completion of the Auto-negotiation process. ANC (bit-5) should be set. The link status (bit-2), when set, indicates that the link is up (only for TBI, SMACMII, and RTBI PHY interfaces).
3. Program the MAC Register 4 (MACMII Address Register) for controlling the management cycles for external PHY. For example, Physical Layer Address PA (bits 15-11). In addition, set bit 0 (MACMII Busy) for writing into PHY and reading from PHY.

4. Read the 16-bit data of Register 5 (MACMII Data Register) from the PHY for link up, speed of operation, and mode of operation, by specifying the appropriate address value in bits 15-11 of Register 4 (MACMII Address Register).
5. Provide the MAC address registers (Register 16 (MAC Address0 High Register) and Register 17 (MAC Address0 Low Register)). If more than one MAC address is enabled in your configuration, program the MAC addresses appropriately.
6. Provide the MAC address registers (Register 16 (MAC Address0 High Register) and Register 17 (MAC Address0 Low Register)). If more than one MAC address is enabled in your configuration, program the MAC addresses appropriately.
7. Program the following fields to set the appropriate filters for the incoming frames in Register 1 (MAC Frame Filter):
 - a. Receive All
 - b. Promiscuous mode
 - c. Hash or Perfect Filter
 - d. Unicast, multicast, broadcast, and control frames filter settings
8. Program the following fields for proper flow control in Register 6 (Flow Control Register):
 - a. Pause time and other pause frame control bits
 - b. Receive and Transmit Flow control bits
 - c. Flow Control Busy/Backpressure Activate
9. Program the Interrupt Mask register bits, as required, and if applicable, for your configuration.
10. Program the appropriate fields in Register 0 (MAC Configuration Register). For example, Inter-frame gap while transmission and jabber disable. Based on the Auto-negotiation you can set the Duplex mode (bit 11) or port select (bit 15).
11. Set the bits Transmit enable (TE bit-3) and Receive Enable (RE bit-2) in Register 0 (MAC Configuration Register).

29.6.3 Performing Normal Receive and Transmit Operation

For normal operation, perform the following steps:

1. For normal transmit and receive interrupts, read the interrupt status. Then, poll the descriptors, reading the status of the descriptor owned by the Host (either transmit or receive).
2. Set appropriate values for the descriptors, ensuring that transmit and receive descriptors are owned by the DMA to resume the transmission and reception of data.
3. If the descriptors are not owned by the DMA (or no descriptor is available), the DMA goes into SUSPEND state. The transmission or reception can be resumed by freeing the descriptors and issuing a poll demand by writing 0 into the Tx/Rx poll demand register (Register 1 (Transmit Poll Demand Register) and Register 2 (Receive Poll Demand Register)).
4. The values of the current host transmitter or receiver descriptor address pointer can be read for the debug process (Register 18 (Current Host Transmit Descriptor Register) and Register 19 (Current Host Receive Descriptor Register)).
5. The values of the current host transmit buffer address pointer and receive buffer address pointer can be read for the debug process (Register 20 (Current Host Transmit Buffer Address Register) and Register 21 (Current Host Receive Buffer Address Register)).

29.6.4 Stopping and Starting Transmission

Perform the following steps to pause the transmission for some time:

1. Disable the Transmit DMA (if applicable), by clearing bit 13 (ST: Start/Stop Transmission Command) of Register 6 (Operation Mode Register).
2. Wait for any previous frame transmissions to complete. You can check this by reading the appropriate bits of Register 9 (Debug Register).
3. Disable the MAC transmitter and MAC receiver by clearing the bit 3 (TE: Transmitter Enable) and bit 2 (RE: Receiver Enable) in Register 0 (MAC Configuration Register).
4. Disable the Receive DMA (if applicable), after making sure that the data in the Rx FIFO is transferred to the system memory (by reading Register 9 (Debug Register)).
5. Make sure that both Tx FIFO and Rx FIFO are empty.
6. To re-start the operation, first start the DMAs, and then enable the MAC Transmitter and Receiver.